

# Program overview

19-May-2025 10:49

**Year** 2024/2025  
**Organization** Civil Engineering and Geosciences  
**Education** Bachelor Civil Engineering

| Code                       | Omschrijving                                        | ECTS | p1 | p2 | p3 | p4 | p5 |
|----------------------------|-----------------------------------------------------|------|----|----|----|----|----|
| <b>CT Bachelor 2024</b>    |                                                     |      |    |    |    |    |    |
| <b>CT BSc 1e jaar 2024</b> |                                                     |      |    |    |    |    |    |
| CTB1000                    | Construction Site 1                                 | 8    |    |    |    |    |    |
| CTB1000-D1                 | Werkboek                                            | 2    |    |    |    |    |    |
| CTB1000-D2                 | Tekenen en planning                                 | 2    |    |    |    |    |    |
| CTB1000-D3                 | Diverse software                                    | 2    |    |    |    |    |    |
| CTB1000-D4                 | CAD/BIM/BEM                                         | 2    |    |    |    |    |    |
| CTB1001-16                 | Analyse                                             | 6    |    |    |    |    |    |
| CTB1002                    | Lineaire Algebra                                    | 6    |    |    |    |    |    |
| CTB1002 Toets 1            | Toets 1                                             | 3    |    |    |    |    |    |
| CTB1002 Toets 2            | Toets 2                                             | 3    |    |    |    |    |    |
| CTB1110-17                 | Construction Mechanics 1                            | 5    |    |    |    |    |    |
| CTB1110-17 Toets 1         | Ethics CTB1110-17                                   | 0    |    |    |    |    |    |
| CTB1110-17 Toets 2         | Structural Mechanics 1                              | 5    |    |    |    |    |    |
| CTB1120-17                 | Introduction to Civil and Environmental Engineering | 5    |    |    |    |    |    |
| CTB1120-17 Toets 1         | Deeltoets 1                                         | 0    |    |    |    |    |    |
| CTB1120-17 Toets 2         | Deeltoets 2                                         | 0    |    |    |    |    |    |
| CTB1120-17 Toets 3         | Ethics CTB1120-17                                   | 0    |    |    |    |    |    |
| CTB1120-17 Toets 4         | Workshops                                           | 0    |    |    |    |    |    |
| CTB1120-17 Toets 5         | Projectanalyse                                      | 2    |    |    |    |    |    |
| CTB1120-17 Toets 6         | Integrale Toets                                     | 3    |    |    |    |    |    |
| CTB1215                    | Clean Watersystems                                  | 5    |    |    |    |    |    |
| CTB1220-17                 | Integrated Design and Control                       | 5    |    |    |    |    |    |
| CTB1220-17 Toets 1         | Deeltoets 1                                         | 0    |    |    |    |    |    |
| CTB1220-17 Toets 2         | Deeltoets 2                                         | 0    |    |    |    |    |    |
| CTB1220-17 Toets 3         | Ethics CTB1220-17                                   | 0    |    |    |    |    |    |
| CTB1220-17 Toets 4         | Project                                             | 3,5  |    |    |    |    |    |
| CTB1220-17 Toets 5         | Integrale Toets                                     | 1,5  |    |    |    |    |    |
| CTB1310                    | Construction Mechanics 2                            | 5    |    |    |    |    |    |
| CTB1320-24                 | Construction Materials and Sustainability           | 5    |    |    |    |    |    |
| CTB1320-24 Toets 1         | Partial exam 1                                      | 0    |    |    |    |    |    |
| CTB1320-24 Toets 2         | Partial exam 2                                      | 0    |    |    |    |    |    |
| CTB1320-24 Toets 3         | Ethics                                              | 0    |    |    |    |    |    |
| CTB1320-24 Toets 4         | Case study                                          | 0    |    |    |    |    |    |
| CTB1320-24 Toets 5         | Integral exam                                       | 5    |    |    |    |    |    |
| CTB1410-20                 | Design of Structures and Foundations 1              | 5    |    |    |    |    |    |
| CTB1410-20 Toets 1         | Constructies en Funderingen                         | 3    |    |    |    |    |    |
| CTB1410-20 Toets 2         | Ontwerppracticum                                    | 2    |    |    |    |    |    |
| CTB1420-17                 | Transport en Planning                               | 5    |    |    |    |    |    |
| CTB1420-17 Toets 1         | Deeltentamen Delen 1 en 2 van het dictaat           | 0    |    |    |    |    |    |
| CTB1420-17 Toets 2         | Deeltentamen Delen 3 en 4 van het dictaat           | 0    |    |    |    |    |    |
| CTB1420-17 Toets 3         | Ethics CTB1420-17                                   | 0    |    |    |    |    |    |
| CTB1420-17 Toets 4         | Oefening CTB1420-17                                 | 1    |    |    |    |    |    |
| CTB1420-17 Toets 5         | Integrale Toets Delen 1 t/m 5                       | 4    |    |    |    |    |    |
| <b>CT BSc 2e jaar 2024</b> |                                                     |      |    |    |    |    |    |
| CTB2000-20                 | Construction Site 2                                 | 8    |    |    |    |    |    |
| CTB2000-20-D1a             | Programmeren 1                                      | 1    |    |    |    |    |    |
| CTB2000-20-D1b             | Vloeistofmechanica practicum                        | 1    |    |    |    |    |    |
| CTB2000-20-D2a             | Programmeren 2                                      | 1    |    |    |    |    |    |
| CTB2000-20-D2b             | Materiaalkunde practicum                            | 1    |    |    |    |    |    |
| CTB2000-20-D3a             | Programmeren 3                                      | 1    |    |    |    |    |    |
| CTB2000-20-D3b             | Grondmechanica practicum                            | 1    |    |    |    |    |    |
| CTB2000-20-D4a             | Programmeren 4                                      | 1    |    |    |    |    |    |
| CTB2000-20-D4b             | Numerieke wiskunde practicum                        | 1    |    |    |    |    |    |
| CTB2105                    | Differentiaalvergelijkingen                         | 3    |    |    |    |    |    |
| CTB2105 Toets 1            | Exam                                                | 3    |    |    |    |    |    |
| CTB2105 Toets 2            | Practical                                           | 0    |    |    |    |    |    |
| CTB2110                    | Fluid Mechanics                                     | 5    |    |    |    |    |    |
| CTB2121                    | Urban Water and Environmental Technology            | 5    |    |    |    |    |    |
| CTB2121 Toets 1            | Deeltentamen Stedelijk Water                        | 0    |    |    |    |    |    |
| CTB2121 Toets 2            | Deeltentamen Milieutechniek                         | 0    |    |    |    |    |    |
| CTB2121 Toets 3            | Integraal tentamen                                  | 3    |    |    |    |    |    |
| CTB2121 Toets 4            | Integrale opdracht                                  | 2    |    |    |    |    |    |

|                                                                                                                 |                                                      |     |  |
|-----------------------------------------------------------------------------------------------------------------|------------------------------------------------------|-----|--|
| CTB2121 Toets 5                                                                                                 | Workshop                                             | 0   |  |
| CTB2200                                                                                                         | Kansrekening & Statistiek                            | 3   |  |
| CTB2210                                                                                                         | Construction Mechanics 3                             | 5   |  |
| CTB2220-14                                                                                                      | Concrete and Steel Structures                        | 5   |  |
| CTB2220-14 Toets 1                                                                                              | Deeltoets 1                                          | 0   |  |
| CTB2220-14 Toets 2                                                                                              | Deeltoets 2                                          | 0   |  |
| CTB2220-14 Toets 3                                                                                              | Oefening                                             | 0   |  |
| CTB2220-14 Toets 4                                                                                              | Integrale Toets                                      | 5   |  |
| CTB2300                                                                                                         | Dynamics of Systems                                  | 3   |  |
| CTB2310                                                                                                         | Soil Mechanics                                       | 5   |  |
| CTB2320-17                                                                                                      | Design of Structures and Foundations 2               | 5   |  |
| CTB2320-17 Toets 1                                                                                              | Deeltoets 1                                          | 0   |  |
| CTB2320-17 Toets 2                                                                                              | Deeltoets 2                                          | 0   |  |
| CTB2320-17 Toets 3                                                                                              | Ethics CTB2320-17                                    | 0   |  |
| CTB2320-17 Toets 4                                                                                              | Practicum                                            | 0   |  |
| CTB2320-17 Toets 5                                                                                              | Integrale Toets                                      | 2   |  |
| CTB2400                                                                                                         | Numerical Mathematics                                | 3   |  |
| CTB2410                                                                                                         | Hydraulic Engineering                                | 5   |  |
| CTB2420-17                                                                                                      | Hydrology                                            | 5   |  |
| CTB2420-17 Toets 1                                                                                              | Deeltoets 1                                          | 0   |  |
| CTB2420-17 Toets 2                                                                                              | Deeltoets 2                                          | 0   |  |
| CTB2420-17 Toets 3                                                                                              | Ethics CTB2420-17                                    | 0   |  |
| CTB2420-17 Toets 4                                                                                              | Practicum                                            | 0   |  |
| CTB2420-17 Toets 5                                                                                              | Integrale Toets                                      | 5   |  |
| <b>CT BSc 3e jaar 2024</b>                                                                                      |                                                      |     |  |
| <b>CT BSc 3e jaar - minor</b>                                                                                   |                                                      |     |  |
| <b>CT BSc 3e jaar - major</b>                                                                                   |                                                      |     |  |
| CTB3000-16                                                                                                      | Bachelor Thesis                                      | 10  |  |
| CTB3000-16 Toets 1                                                                                              | Ethics                                               | 0   |  |
| CTB3000-16 Toets 2                                                                                              | Presentation                                         | 10  |  |
| CTB3310                                                                                                         | Surveying and Mapping                                | 4   |  |
| CTB3310 Toets 1                                                                                                 | Practicum                                            | 0   |  |
| CTB3310 Toets 2                                                                                                 | Tentamen                                             | 4   |  |
| CTB3320                                                                                                         | Road and Railway Design                              | 4   |  |
| <b>Overzicht Specialisatievakken. Student dient drie specialisatievakken te kiezen uit de lijsten hieronder</b> |                                                      |     |  |
| <b>Sterk aanbevolen als voorbereiding op de MSc Environmental Engineering, kies 3 uit:</b>                      |                                                      |     |  |
| CTB3311                                                                                                         | Climate Impacts and Engineering                      | 4   |  |
| CTB3311 Toets 1                                                                                                 | Exam                                                 | 2,5 |  |
| CTB3311 Toets 2                                                                                                 | Assignment                                           | 1,5 |  |
| CTB3350                                                                                                         | Open Channel Flow                                    | 4   |  |
| CTB3360                                                                                                         | Water System Analysis                                | 4   |  |
| CTB3365-16                                                                                                      | Introduction to Water Treatment                      | 4   |  |
| CTB3390                                                                                                         | Mechanics and Transport by Flow through Poreus Media | 4   |  |
| CTB3415                                                                                                         | Water Management Research                            | 4   |  |
| <b>Sterk aanbevolen als voorbereiding op de MSc Civil Engineering, kies 3 uit:</b>                              |                                                      |     |  |
| CTB3330                                                                                                         | Structural Mechanics 4                               | 4   |  |
| CTB3335                                                                                                         | Concrete Structures 2                                | 4   |  |
| CTB3340-15                                                                                                      | Building Structures 1                                | 4   |  |
| CTB3350                                                                                                         | Open Channel Flow                                    | 4   |  |
| CTB3355                                                                                                         | Hydraulic Structures                                 | 4   |  |
| CTB3370-18                                                                                                      | Geometrical Design of Roads and Railways             | 4   |  |
| CTB3385                                                                                                         | Use of Underground Space                             | 4   |  |
| CTB3390                                                                                                         | Mechanics and Transport by Flow through Poreus Media | 4   |  |
| CTB3420                                                                                                         | Integral Design of Infrastructure                    | 4   |  |
| CTB3425-17                                                                                                      | Monitoring and Stability of Dikes and Embankments    | 4   |  |

|                     |                                          |
|---------------------|------------------------------------------|
| <b>Year</b>         | <b>2024/2025</b>                         |
| <b>Organization</b> | <b>Civil Engineering and Geosciences</b> |
| <b>Education</b>    | <b>Bachelor Civil Engineering</b>        |

CT Bachelor 2024

|                     |                                          |
|---------------------|------------------------------------------|
| <b>Year</b>         | <b>2024/2025</b>                         |
| <b>Organization</b> | <b>Civil Engineering and Geosciences</b> |
| <b>Education</b>    | <b>Bachelor Civil Engineering</b>        |

**CT BSc 1e jaar 2024**

| CTB1000                                 | Construction Site 1                                                                                                                                                                                                                                                                                                                        | 8 |
|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>           | Ir. S. Pasterkamp                                                                                                                                                                                                                                                                                                                          |   |
| <b>Contact Hours / Week</b><br>x/x/x/x  | 4/4/4/4                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Education Period</b>                 | 1<br>2<br>3<br>4                                                                                                                                                                                                                                                                                                                           |   |
| <b>Start Education</b>                  | 1<br>2<br>3<br>4                                                                                                                                                                                                                                                                                                                           |   |
| <b>Exam Period</b>                      | none                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Course Language</b>                  | Dutch                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Course Contents</b>                  | Zie deelvakken CTB1000-D1, CTB1000-D2, CTB1000-D3, CTB1000-D4                                                                                                                                                                                                                                                                              |   |
| <b>Study Goals</b>                      | Zie deelvakken CTB1000-D1, CTB1000-D2, CTB1000-D3, CTB1000-D4                                                                                                                                                                                                                                                                              |   |
| <b>Education Method</b>                 | Zie deelvakken CTB1000-D1, CTB1000-D2, CTB1000-D3, CTB1000-D4                                                                                                                                                                                                                                                                              |   |
| <b>Course Relations</b>                 | Elk deelvak van de Bouwplaats heeft relaties met andere vakken die tegelijkertijd met dat deelvak gegeven worden. Er wordt samengewerkt met name op de onderwerpen (bijvoorbeeld een gebouw, een brug) waarmee geoefend wordt. Een constructie kan bijvoorbeeld bij Constructief Ontwerpen ontworpen, en in de Bouwplaats getekend worden. |   |
| <b>Assessment</b>                       | Zie deelvakken CTB1000-D1, CTB1000-D2, CTB1000-D3, CTB1000-D4                                                                                                                                                                                                                                                                              |   |
| <b>Tags</b>                             | Design<br>Drawing<br>Group work<br>Intensive<br>Modelling<br>Practicals<br>Process<br>Project<br>Software                                                                                                                                                                                                                                  |   |
| <b>Expected prior Knowledge</b>         | None                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Academic Skills</b>                  | Zie deelvakken CTB1000-D1, CTB1000-D2, CTB1000-D3, CTB1000-D4                                                                                                                                                                                                                                                                              |   |
| <b>Literature &amp; Study Materials</b> | Zie deelvakken CTB1000-D1, CTB1000-D2, CTB1000-D3, CTB1000-D4                                                                                                                                                                                                                                                                              |   |
| <b>Judgement</b>                        | Zie deelvakken CTB1000-D1, CTB1000-D2, CTB1000-D3, CTB1000-D4                                                                                                                                                                                                                                                                              |   |
| <b>Permitted Materials during Exam</b>  | N/A                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Collegerama</b>                      | No                                                                                                                                                                                                                                                                                                                                         |   |

| CTB1000-D1                              | Werkboek                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2 |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>           | Ir. S. Pasterkamp                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Contact Hours / Week</b><br>x/x/x/x  | 4/0/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Education Period</b>                 | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Start Education</b>                  | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Exam Period</b>                      | 1<br>2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Course Language</b>                  | Dutch                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Parts</b>                            | Practicummiddagen waarin de verschillende vaardigheden worden bijgebracht door het maken van oefeningen.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Summary</b>                          | Het voornaamste doel van dit vak is om te leren:<br>Rapporteren, dataverzameling, tekenen                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Course Contents</b>                  | Rapportagetechniek<br>Brightspace course Information Literacy 1<br>Handschetsen<br>Technisch tekenen                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Study Goals</b>                      | <p>M.b.t. rapporteren kan de student:</p> <ul style="list-style-type: none"> <li>Een effectief technisch rapport schrijven;</li> <li>Een effectieve structuur voor een technisch rapport opzetten;</li> <li>Een effectieve introductie en samenvatting schrijven;</li> <li>Overtuigende conclusies en aanbevelingen schrijven;</li> <li>Een duidelijke en aantrekkelijke schrijfstijl gebruiken;</li> <li>Functionele illustraties en tabellen gebruiken.</li> </ul> <p>M.b.t. de kennisverwerving kan de student:</p> <ul style="list-style-type: none"> <li>de kennisbehoefte vaststellen.</li> <li>gericht zoeken door middel van een zoekplan.</li> <li>de gevonden informatie beoordelen op bruikbaarheid en betrouwbaarheid.</li> <li>informatie bronnen weergeven in een database en in groepsverband beheren.</li> </ul> <p>M.b.t. tekenen kan de student:</p> <ul style="list-style-type: none"> <li>Het vaardig gebruiken van potlood, pennen, stiften, liniaal, tekenpapier, schetsrol</li> <li>Kijken en vertalen naar papier: het snel kunnen omzetten van een object dat je in het echt ziet (bijv. gebouw, brug, weg, sluis, viaduct, constructie of onderdelen hiervan) in een 2D of 3D handschets</li> <li>Ontwerpen: het snel kunnen maken van een 2D of 3D handschets van (een onderdeel van) een zelf bedachte technische oplossing</li> <li>Het hebben dan wel ontwikkelen van ruimtelijk inzicht</li> <li>Het kunnen communiceren met anderen op basis van schetsen en uitwerking hiervan in een technische tekening</li> <li>Begrijpen en toepassen van de Amerikaanse projectie;</li> <li>Ruimtelijk inzicht en het vertalen van 3D naar 2D;</li> <li>Opbouw van een technische tekening;</li> <li>Maken van een eenvoudige technische tekening;</li> <li>Toepassen van de tekenregels;</li> <li>Aanduiding van details in tekeningen;</li> <li>Intekenen van betonwapening;</li> <li>Nieuwe vormen van maataanduiding;</li> <li>Maataanduidingen door middel van bijschriften;</li> <li>De opbouw van de hoofddragconstructie van een eenvoudige hal begrijpen.</li> </ul> <p>Studielast:<br/>Ca. 4 u/w begeleid, 4 u/w zelfstudie</p> |   |
| <b>Education Method</b>                 | Opdrachtmiddagen, zelfstudie                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Course Relations</b>                 | Inleiding civiele techniek                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Assessment</b>                       | De leerdoelen worden getoetst door middel van:<br>Opdrachten<br>Alle opdrachten moeten als voldoende beoordeeld zijn.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Tags</b>                             | Drawing                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Expected prior Knowledge</b>         | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Academic Skills</b>                  | Drawing, academic writing, referencing your work, preventing plagiarism, finding information in a library                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Literature &amp; Study Materials</b> | Rapportagetechniek, 7e druk 2023, verkrijgbaar bij Waltmans Bouwshop<br>Tekenset, verkrijgbaar bij Waltmans Bouwshop<br>Dictaat CTB1000 Technisch Tekenin Versie Juli 2013, verkrijgbaar via Microweb                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Judgement</b>                        | Voor informatie over de beoordeling zie artikel 14 van de Regels en Richtlijnen (RRvE), alsmede artikel 9 van de bijlage bij de Onderwijs- en Examenregeling (OER):<br><a href="https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations">https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Permitted Materials during Exam</b>  | N/A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Collegerama</b>                      | No                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |

| CTB1000-D2                              | Tekenen en planning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2 |
|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>           | Ir. S. Pasterkamp                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Contact Hours / Week</b><br>x/x/x/x  | 0/4/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Education Period</b>                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Start Education</b>                  | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Exam Period</b>                      | 2<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Course Language</b>                  | Dutch                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Parts</b>                            | Practicummiddagen waarin de verschillende vaardigheden worden bijgebracht door het maken van oefeningen.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Summary</b>                          | Het voornaamste doel van dit vak is om te leren:<br>Projectvaardigheden                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Course Contents</b>                  | Technisch handtekenen<br>Eenvoudig CAD-tekenen (Rhino)<br>Opstellen (financiële) planning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Study Goals</b>                      | De fasering van een project kunnen benoemen;<br>Samenwerken in een projectgroep;<br>Resultaten presenteren;<br>Aan te geven waar de eigen voorkeuren en kwaliteiten liggen m.b.t. de werkzaamheden in de verschillende fasen;<br>de inpassing van subsystemen (bouwwerken) in een omgeving vaststellen en weergeven;<br>de vorm van een ontwerp vertalen in de werking (hetgeen er wordt bewerkstelligd), de wijze van opbouw, werking en capaciteiten;<br>de verschillen zien tussen een systeem, subsysteem en sub-subsysteem (van grof naar fijn);<br>de werkelijkheid kritisch opnemen en weergeven;<br>Het effect van het gebruik op de omgeving weergeven en dit vertalen in ofwel een knelpunt of randvoorwaarden voor toekomstige plannen;<br>Herkennen eigen voorkeur qua rol in een project;<br>Reflecteren op individuele inbreng in groepsproces;<br>Project op systeem- en subsysteemniveau in kaart brengen aan de hand van plattegronden, schetsen en tekeningen;<br>Een ontwerp inpassen in de omgeving;<br>Een besluitvormingsproces in een groep doorgronden en beïnvloeden;<br>Het opstellen van een Work breakdown Structure (WBS);<br>Het maken van een projectplanning dmv planningssoftware;<br><br>Rhino<br>basisvaardigheden Rhino;<br>3D schetsen met de computer;<br>presentatie-afbeeldingen maken.<br><br>Studielast:<br>Ca. 4 u/w begeleid, 4 u/w zelfstudie |   |
| <b>Education Method</b>                 | Opdrachtmiddagen                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Course Relations</b>                 | Integraal Ontwerpen                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Assessment</b>                       | De leerdoelen worden getoetst door middel van:<br>Opdrachten<br>Alle opdrachten moeten als voldoende beoordeeld zijn.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Tags</b>                             | Drawing<br>Personal<br>Software                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Expected prior Knowledge</b>         | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Academic Skills</b>                  | drawing, working in groups, finding information in a library                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Literature &amp; Study Materials</b> | Alle lesmateriaal wordt gepresenteerd via Brightspace.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Judgement</b>                        | Voor informatie over de beoordeling zie artikel 14 van de Regels en Richtlijnen (RRvE), alsmede artikel 9 van de bijlage bij de Onderwijs- en Examenregeling (OER):<br><a href="https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations">https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Permitted Materials during Exam</b>  | N/A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Collegerama</b>                      | No                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |

| <b>CTB1000-D3</b>                       | <b><i>Diverse software</i></b>                                                                                                                                                                                                                                                                                                                                                                                                     | <b>2</b> |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| <b>Responsible Instructor</b>           | Ir. S. Pasterkamp                                                                                                                                                                                                                                                                                                                                                                                                                  |          |
| <b>Contact Hours / Week</b><br>x/x/x/x  | 0/0/4/0                                                                                                                                                                                                                                                                                                                                                                                                                            |          |
| <b>Education Period</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                  |          |
| <b>Start Education</b>                  | 3                                                                                                                                                                                                                                                                                                                                                                                                                                  |          |
| <b>Exam Period</b>                      | 3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                             |          |
| <b>Course Language</b>                  | Dutch                                                                                                                                                                                                                                                                                                                                                                                                                              |          |
| <b>Required for</b>                     | CTB1000-D4                                                                                                                                                                                                                                                                                                                                                                                                                         |          |
| <b>Course Contents</b>                  | BIM, Tekla, Revit                                                                                                                                                                                                                                                                                                                                                                                                                  |          |
| <b>Study Goals</b>                      | Weten wat BIM is;<br>Weten wat er nodig is om BIM te kunnen toepassen bij een bouwproject;<br>Het bekend zijn met de (on)mogelijkheden van BIM / BEM<br>Een constructie kunnen modelleren met Tekla;<br>Een constructie kunnen modelleren met Revit;<br>Een materialenstaat kunnen uittrekken uit een BIM-model;<br>Op efficiënte wijze technische tekeningen en presentatietekeningen kunnen maken vanuit een BIM-model.          |          |
| <b>Education Method</b>                 | Opdrachtmiddagen, zelfstudie                                                                                                                                                                                                                                                                                                                                                                                                       |          |
| <b>Course Relations</b>                 | CTB1320-17                                                                                                                                                                                                                                                                                                                                                                                                                         |          |
| <b>Assessment</b>                       | De leerdoelen worden getoetst door middel van:<br>Opdrachten<br>Alle opdrachten moeten als voldoende beoordeeld zijn.                                                                                                                                                                                                                                                                                                              |          |
| <b>Tags</b>                             | Drawing<br>Software                                                                                                                                                                                                                                                                                                                                                                                                                |          |
| <b>Expected prior Knowledge</b>         | CTB1000-D1                                                                                                                                                                                                                                                                                                                                                                                                                         |          |
| <b>Academic Skills</b>                  | Digital modelling                                                                                                                                                                                                                                                                                                                                                                                                                  |          |
| <b>Literature &amp; Study Materials</b> | Alle lesmateriaal wordt gepresenteerd via Brightspace.                                                                                                                                                                                                                                                                                                                                                                             |          |
| <b>Judgement</b>                        | Voor informatie over de beoordeling zie artikel 14 van de Regels en Richtlijnen (RRvE), alsmede artikel 9 van de bijlage bij de Onderwijs- en Examenregeling (OER):<br><a href="https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations">https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations</a> |          |
| <b>Permitted Materials during Exam</b>  | N/A                                                                                                                                                                                                                                                                                                                                                                                                                                |          |
| <b>Collegerama</b>                      | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                |          |

| <b>CTB1000-D4</b>                       | <b><i>CAD/BIM/BEM</i></b>                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>2</b> |
|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| <b>Responsible Instructor</b>           | Ir. S. Pasterkamp                                                                                                                                                                                                                                                                                                                                                                                                                                |          |
| <b>Contact Hours / Week</b><br>x/x/x/x  | 0/0/0/4                                                                                                                                                                                                                                                                                                                                                                                                                                          |          |
| <b>Education Period</b>                 | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                |          |
| <b>Start Education</b>                  | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                |          |
| <b>Exam Period</b>                      | 4<br>5                                                                                                                                                                                                                                                                                                                                                                                                                                           |          |
| <b>Course Language</b>                  | Dutch                                                                                                                                                                                                                                                                                                                                                                                                                                            |          |
| <b>Course Contents</b>                  | BIM                                                                                                                                                                                                                                                                                                                                                                                                                                              |          |
| <b>Study Goals</b>                      | Het begrijpen van de consequenties van het gebruik van BIM / BEM voor het ontwerpproces<br>Het binnen een groep ontwerpers op effectieve wijze kunnen maken van afspraken over versies, mutaties, bevoegdheden, beheer, etc.<br>Modelinformatie kunnen uitwisselen tussen verschillende modelleerpakketten;<br>Clashcontrole kunnen uitvoeren;<br>Het kunnen uitwerken en visualiseren van eenvoudige 4D of 5D informatie op grond van het model |          |
| <b>Education Method</b>                 | Opdrachtmiddagen, zelfstudie                                                                                                                                                                                                                                                                                                                                                                                                                     |          |
| <b>Course Relations</b>                 | CTB1410-20                                                                                                                                                                                                                                                                                                                                                                                                                                       |          |
| <b>Assessment</b>                       | De leerdoelen worden getoetst door middel van:<br>Opdrachten<br>Alle opdrachten moeten als voldoende beoordeeld zijn.                                                                                                                                                                                                                                                                                                                            |          |
| <b>Tags</b>                             | Design<br>Group work<br>Modelling<br>Practicals<br>Software                                                                                                                                                                                                                                                                                                                                                                                      |          |
| <b>Expected prior Knowledge</b>         | CTB1000-D3                                                                                                                                                                                                                                                                                                                                                                                                                                       |          |
| <b>Academic Skills</b>                  | Digital modelling, learning how to work with new digital tools, working in groups, oral presentation                                                                                                                                                                                                                                                                                                                                             |          |
| <b>Literature &amp; Study Materials</b> | Alle lesmateriaal wordt gepresenteerd via Brightspace.                                                                                                                                                                                                                                                                                                                                                                                           |          |
| <b>Judgement</b>                        | Voor informatie over de beoordeling zie artikel 14 van de Regels en Richtlijnen (RRvE), alsmede artikel 9 van de bijlage bij de Onderwijs- en Examenregeling (OER):<br><a href="https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations">https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations</a>               |          |
| <b>Permitted Materials during Exam</b>  | N/A                                                                                                                                                                                                                                                                                                                                                                                                                                              |          |
| <b>Collegerama</b>                      | No                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |



| CTB1001-16                     | Analyse                           | 6 |
|--------------------------------|-----------------------------------|---|
| Responsible Instructor         | Dr.ir. N.P.K. Chan                |   |
| Co-responsible for assignments | Dr.ir. D. den Ouden-van der Horst |   |
| Education Period               | 1<br>2                            |   |
| Start Education                | 1<br>2                            |   |
| Exam Period                    | 1<br>2<br>3                       |   |
| Course Language                | Dutch                             |   |
| Collegerama                    | No                                |   |

| CTB1002                        | Lineaire Algebra                  | 6 |
|--------------------------------|-----------------------------------|---|
| Responsible Instructor         | Dr. O. Jaibi                      |   |
| Co-responsible for assignments | Dr.ir. D. den Ouden-van der Horst |   |
| Education Period               | 3<br>4                            |   |
| Start Education                | 3                                 |   |
| Exam Period                    | 3<br>4<br>5                       |   |
| Course Language                | Dutch                             |   |
| Collegerama                    | No                                |   |

| CTB1002 Toets 1        | Toets 1      | 3 |
|------------------------|--------------|---|
| Responsible Instructor | Dr. O. Jaibi |   |
| Education Period       | 3            |   |
| Start Education        | 3            |   |
| Exam Period            | 3<br>4       |   |
| Course Language        | Dutch        |   |
| Collegerama            | No           |   |

| CTB1002 Toets 2        | Toets 2      | 3 |
|------------------------|--------------|---|
| Responsible Instructor | Dr. O. Jaibi |   |
| Education Period       | 4            |   |
| Start Education        | 4            |   |
| Exam Period            | 4<br>5       |   |
| Course Language        | Dutch        |   |
| Collegerama            | No           |   |

| CTB1110-17             | Construction Mechanics 1                                  | 5 |
|------------------------|-----------------------------------------------------------|---|
| Responsible Instructor | Ir. J.W. Welleman                                         |   |
| Instructor             | T.R. van Woudenberg                                       |   |
| Education Period       | 1                                                         |   |
| Start Education        | 1                                                         |   |
| Exam Period            | 1<br>2                                                    |   |
| Course Language        | Dutch                                                     |   |
| Tags                   | Algebra<br>Analysis<br>Calculus<br>Mechanics<br>Modelling |   |
| Collegerama            | No                                                        |   |

|                           |                          |          |
|---------------------------|--------------------------|----------|
| <b>CTB1110-17 Toets 1</b> | <b>Ethics CTB1110-17</b> | <b>0</b> |
| Responsible Instructor    | Ir. J.W. Welleman        |          |
| Education Period          | 1                        |          |
| Start Education           | 1                        |          |
| Exam Period               | 1<br>2                   |          |
| Course Language           | Dutch                    |          |
| Collegerama               | No                       |          |

|                           |                               |          |
|---------------------------|-------------------------------|----------|
| <b>CTB1110-17 Toets 2</b> | <b>Structural Mechanics 1</b> | <b>5</b> |
| Responsible Instructor    | Ir. J.W. Welleman             |          |
| Education Period          | 1                             |          |
| Start Education           | 1                             |          |
| Exam Period               | 1<br>2                        |          |
| Course Language           | Dutch                         |          |
| Collegerama               | No                            |          |

|                        |                                                            |          |
|------------------------|------------------------------------------------------------|----------|
| <b>CTB1120-17</b>      | <b>Introduction to Civil and Environmental Engineering</b> | <b>5</b> |
| Responsible Instructor | Prof.dr.ir. L.C. Rietveld                                  |          |
| Education Period       | 1                                                          |          |
| Start Education        | 1                                                          |          |
| Exam Period            | 1                                                          |          |
| Course Language        | Dutch                                                      |          |
| Collegerama            | No                                                         |          |

|                           |                           |          |
|---------------------------|---------------------------|----------|
| <b>CTB1120-17 Toets 1</b> | <b>Deeltoets 1</b>        | <b>0</b> |
| Responsible Instructor    | Prof.dr.ir. L.C. Rietveld |          |
| Education Period          | 1                         |          |
| Start Education           | 1                         |          |
| Exam Period               | 1                         |          |
| Course Language           | Dutch                     |          |
| Collegerama               | No                        |          |

|                           |                           |          |
|---------------------------|---------------------------|----------|
| <b>CTB1120-17 Toets 2</b> | <b>Deeltoets 2</b>        | <b>0</b> |
| Responsible Instructor    | Prof.dr.ir. L.C. Rietveld |          |
| Education Period          | 1                         |          |
| Start Education           | 1                         |          |
| Exam Period               | 1                         |          |
| Course Language           | Dutch                     |          |
| Collegerama               | No                        |          |

|                           |                           |          |
|---------------------------|---------------------------|----------|
| <b>CTB1120-17 Toets 3</b> | <b>Ethics CTB1120-17</b>  | <b>0</b> |
| Responsible Instructor    | Prof.dr.ir. L.C. Rietveld |          |
| Education Period          | 1                         |          |
| Start Education           | 1                         |          |
| Exam Period               | 1                         |          |
| Course Language           | Dutch                     |          |
| Collegerama               | No                        |          |

|                           |                           |          |
|---------------------------|---------------------------|----------|
| <b>CTB1120-17 Toets 4</b> | <b>Workshops</b>          | <b>0</b> |
| Responsible Instructor    | Prof.dr.ir. L.C. Rietveld |          |
| Education Period          | 1                         |          |
| Start Education           | 1                         |          |
| Exam Period               | 1                         |          |
| Course Language           | Dutch                     |          |
| Collegerama               | No                        |          |

| <b>CTB1120-17 Toets 5</b>     | <b><i>Projectanalyse</i></b> | <b>2</b> |
|-------------------------------|------------------------------|----------|
| <b>Responsible Instructor</b> | Prof.dr.ir. L.C. Rietveld    |          |
| <b>Education Period</b>       | 1                            |          |
| <b>Start Education</b>        | 1                            |          |
| <b>Exam Period</b>            | 1                            |          |
| <b>Course Language</b>        | Dutch                        |          |
| <b>Collegerama</b>            | No                           |          |

| <b>CTB1120-17 Toets 6</b>     | <b><i>Integrale Toets</i></b> | <b>3</b> |
|-------------------------------|-------------------------------|----------|
| <b>Responsible Instructor</b> | Prof.dr.ir. L.C. Rietveld     |          |
| <b>Education Period</b>       | 1                             |          |
| <b>Start Education</b>        | 1                             |          |
| <b>Exam Period</b>            | 1                             |          |
| <b>Course Language</b>        | Dutch                         |          |
| <b>Collegerama</b>            | No                            |          |

| <b>CTB1215</b>                | <b>Clean Watersystems</b>                                                                                                                                                                                                                                                                                                                                              | <b>5</b> |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| <b>Responsible Instructor</b> | Prof.dr.ir. D. van Halem                                                                                                                                                                                                                                                                                                                                               |          |
| <b>Instructor</b>             | Prof.dr.ir. L.C. Rietveld                                                                                                                                                                                                                                                                                                                                              |          |
| <b>Instructor</b>             | Prof.dr.ir. M.K. de Kreuk                                                                                                                                                                                                                                                                                                                                              |          |
| <b>Instructor</b>             | Dr.ir. R.W. Hut                                                                                                                                                                                                                                                                                                                                                        |          |
| <b>Instructor</b>             | Prof.dr.ir. J.B. van Lier                                                                                                                                                                                                                                                                                                                                              |          |
| <b>Education Period</b>       | 2                                                                                                                                                                                                                                                                                                                                                                      |          |
| <b>Start Education</b>        | 2                                                                                                                                                                                                                                                                                                                                                                      |          |
| <b>Exam Period</b>            | 2<br>3                                                                                                                                                                                                                                                                                                                                                                 |          |
| <b>Course Language</b>        | Dutch                                                                                                                                                                                                                                                                                                                                                                  |          |
| <b>Assessment</b>             |                                                                                                                                                                                                                                                                                                                                                                        |          |
| <b>Judgement</b>              | For more information on grading, see article 14 in the Rules and Guidelines (RGBE):<br><a href="https://www.tudelft.nl/en/student/faculties/ceg-student-portal/education/education-information/educational-rules-and-regulations">https://www.tudelft.nl/en/student/faculties/ceg-student-portal/education/education-information/educational-rules-and-regulations</a> |          |

| <b>CTB1220-17</b>             | <b>Integrated Design and Control</b> | <b>5</b> |
|-------------------------------|--------------------------------------|----------|
| <b>Responsible Instructor</b> | Dr. E.J. Houwing                     |          |
| <b>Instructor</b>             | Prof.dr.ir. M.J.C.M. Hertogh         |          |
| <b>Education Period</b>       | 2                                    |          |
| <b>Start Education</b>        | 2                                    |          |
| <b>Exam Period</b>            | 2                                    |          |
| <b>Course Language</b>        | Dutch                                |          |
| <b>Collegerama</b>            | Yes                                  |          |

| <b>CTB1220-17 Toets 1</b>     | <b><i>Deeltoets 1</i></b> | <b>0</b> |
|-------------------------------|---------------------------|----------|
| <b>Responsible Instructor</b> | Dr. E.J. Houwing          |          |
| <b>Education Period</b>       | 2                         |          |
| <b>Start Education</b>        | 2                         |          |
| <b>Exam Period</b>            | 2                         |          |
| <b>Course Language</b>        | Dutch                     |          |
| <b>Collegerama</b>            | Yes                       |          |

| <b>CTB1220-17 Toets 2</b>     | <b><i>Deeltoets 2</i></b> | <b>0</b> |
|-------------------------------|---------------------------|----------|
| <b>Responsible Instructor</b> | Dr. E.J. Houwing          |          |
| <b>Education Period</b>       | 2                         |          |
| <b>Start Education</b>        | 2                         |          |
| <b>Exam Period</b>            | 2                         |          |
| <b>Course Language</b>        | Dutch                     |          |
| <b>Collegerama</b>            | Yes                       |          |

| <b>CTB1220-17 Toets 3</b> | <b><i>Ethics CTB1220-17</i></b> | <b>0</b> |
|---------------------------|---------------------------------|----------|
| Responsible Instructor    | Dr. E.J. Houwing                |          |
| Education Period          | 2                               |          |
| Start Education           | 2                               |          |
| Exam Period               | 2                               |          |
| Course Language           | Dutch                           |          |
| Collegerama               | Yes                             |          |

| <b>CTB1220-17 Toets 4</b> | <b><i>Project</i></b> | <b>3.5</b> |
|---------------------------|-----------------------|------------|
| Responsible Instructor    | Dr. E.J. Houwing      |            |
| Education Period          | 2                     |            |
| Start Education           | 2                     |            |
| Exam Period               | 2                     |            |
| Course Language           | Dutch                 |            |
| Collegerama               | Yes                   |            |

| <b>CTB1220-17 Toets 5</b> | <b><i>Integrale Toets</i></b> | <b>1.5</b> |
|---------------------------|-------------------------------|------------|
| Responsible Instructor    | Dr. E.J. Houwing              |            |
| Education Period          | 2                             |            |
| Start Education           | 2                             |            |
| Exam Period               | 2                             |            |
| Course Language           | Dutch                         |            |
| Collegerama               | Yes                           |            |

| <b>CTB1310</b>         | <b>Construction Mechanics 2</b> | <b>5</b> |
|------------------------|---------------------------------|----------|
| Responsible Instructor | Prof.dr.ir. J.G. Rots           |          |
| Education Period       | 3                               |          |
| Start Education        | 3                               |          |
| Exam Period            | 3<br>4                          |          |
| Course Language        | Dutch                           |          |
| Collegerama            | Yes                             |          |

| CTB1320-24                             | Construction Materials and Sustainability                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 5 |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr. O. Copuroglu                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Instructor</b>                      | Prof.dr. H.M. Jonkers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/6/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Education Period</b>                | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Start Education</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Exam Period</b>                     | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Course Language</b>                 | Dutch<br>English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Course Contents</b>                 | <p>Part I - Construction Materials: This course provides an introductory overview of the general material science and various materials used in construction. The main objective of this course is to provide the students with fundamental knowledge and understanding of how materials are built-up and what are the relationship between important chemical/physical properties and material behavior.</p> <p>Part I - Construction Materials lectures are given in English</p> <p>Part II - Sustainability:<br/>In this part of the course the interaction between materials and the environment, and the resulting consequences for both, will be treated. Production and use of materials go at the expense of finite resources and production of environmental harmful compounds while at the same time environmental conditions affect materials durability and functionality aspects. Focus will be on the quantification of Sustainability using Life Cycle Assessment (LCA) methods.</p> <p>Part II - Sustainability lectures are given in Dutch</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Study Goals</b>                     | <p>ART I - Construction Materials</p> <p>Define material science and material engineering.<br/>Classify main construction materials.<br/>Tell importance of standards and standardization in material production, selection and use<br/>Define and illustrate the basic properties of atoms, different atomic bonds, crystal structures and dislocations. Relate bonding properties to the physical properties of materials.<br/>Relate chemistry to mechanics<br/>Define stress and strain, classify different material behavior under stress, define Poissons ratio. Classify and analyze E-modulus on stress-strain diagrams<br/>Differentiate elastic and plastic behavior. Define and compute tensile and compressive stress, list effect of testing parameters, define material hardness.<br/>Define and explain simple fracture, brittle fracture, and ductile fracture, explain fractography, surface energy and strain energy, derive the Griffiths equation. Define and formulate stress concentrations, fracture toughness, fracture modes. Discuss plain fracture design strategy, fracture toughness testing, and fatigue.<br/>Tell importance of phase diagrams, relate to microstructure and mechanical properties of materials. Define and apply lever rule.<br/>Define and classify ferrous alloys. Explain iron, important iron ores and their chemical properties. Define steel, compare steel with other iron alloys. Illustrate iron-iron carbide systems. Explain various Fe-C microstructures. Explain the effect of carbon content on the stress-strain behavior of steel.<br/>Define rock, tell importance of rocks in civil engineering. Define and discuss the difference between rocks and aggregate.<br/>Define gradation, illustrate grading curve. Discuss importance of grading, define fineness modulus, max. aggregate size and other physical properties of aggregates.<br/>Define portland cement, compare hydraulic and nonhydraulic binders, define and compare paste, mortar and aggregate.<br/>Summarize characteristics of main cement phases. Define cement hydration, hydration reactions, and heat of hydration, Discuss effect of cement minerals on the strength development of concrete.<br/>List components of modern concrete. Classify concrete admixtures. Define and discuss concrete technology, fresh and hardened concrete, mix design, classify of concrete based on weight and strength.<br/>List properties of hardened concrete, explain concrete strength and relate to W/C ratio and degree of hydration. Define W/C and discuss its importance for quality of concrete. Illustrate W/C vs. Strength and W/C vs. capillary porosity.<br/>Explain the concept of durability of concrete. Compare durability with sustainability.<br/>Explain and discuss mechanisms and characteristics of main durability problems.<br/>Design durable concrete based on accumulated knowledge.<br/>Define types and characteristics of structural wood and timber. Explain mechanical properties and durability issues related to wood and timber.</p> <p>PART II - Sustainability<br/>Define, characterize and analyse sustainable construction materials<br/>Quantification of Sustainability: Master the application of Life Cycle Assessment (LCA) methods<br/>Analyse environmental element cycling and interaction with construction materials<br/>Classify environmental aspects of raw material use and emissions of harmful substances<br/>Explain in sustainability context choices between application of finite and renewable raw materials for civil engineering practices<br/>Discuss examples of research concerning development of innovative bio-based sustainable materials<br/>Ethics aspects of environmental and materials engineering</p> <p>Study load:<br/>PART I:<br/>Lecture hours: 24hrs<br/>Self study: 56hrs</p> <p>PART II:<br/>Lecture hours: 16hrs<br/>Case study: 14 hrs<br/>Self study: 30 hrs</p> |   |
| <b>Education Method</b>                | Lecture hours and self study                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Course Relations</b>                | <p>This course forms the fundamentals of the following courses:<br/>All civil engineering materials related courses in the M.Sc. program.</p> <p>Bouwplaats</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Assessment</b>                      | <p>The final grade will be calculated from:<br/>2x examinations (for Construction Materials and Sustainability)<br/>The examinations will be carried out in computer-based environment as long as sufficient facilities can be arranged for the entire number of students. Lecturers will announce the examination style well before the first midterm examination. The exam questions will be offered bilingual, i.e. in English as well as in Dutch language.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |

|                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Expected prior Knowledge</b>         | Recovery (integral) examination covers the entire course content. Midterm exams have a duration of 90 minutes. The integral exam has a duration of 180 minutes.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                                         | A case study assignment (pass/fail) is used for the assessment of the sustainability part.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                         | If a student fails the case study, he/she may not pass the course irrespective to his/her exam grades.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                                         | VWO level chemistry and physics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Academic Skills</b>                  | conceptual thinking, research, reporting, reasoning, problem solving                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Literature &amp; Study Materials</b> | for PART I - Construction Materials:<br><br>Materials Science for Civil Engineering - Lecture Notes by Dr. O. Copuroglu<br><br>Additional reading material for further study (available at the library):<br><br>Any fundamental material science textbook such as:<br><br>- Callister, William D., and David G. Rethwisch. Fundamentals of materials science and engineering. Wiley, 2013.<br><br>- Ashby, Michael F., Hugh Shercliff, and David Cebon. Materials: engineering, science, processing and design. Butterworth-Heinemann, 2013.<br><br>- Materials Science in Design and Engineering, P. van Mourik, J. van Dam, S. Picken, 2012, VSSD publishing. (Also available in Dutch)<br><br>also any textbook on materials of construction is recommended:<br>- Construction Materials: Their Nature and Behaviour, 3rd ed. Edited by J . M . Illston and P . L . J . Spon Press 2001, Print ISBN: 978-0-419-25860-5 eBook ISBN: 978-0-203-47898-1 3rd ed. is free for the TU Delft students at:<br><a href="http://www.crcnetbase.com/isbn/978-0-419-25860-5">http://www.crcnetbase.com/isbn/978-0-419-25860-5</a><br><br>The following ebook is obligatory for PART II - Sustainability:<br>Reader CTB1320 Sustainability<br><br>For both parts:<br>PDF files of the lecture slides and homework exercises<br><br>Final grade = 1st partial examination (50%) + 2nd partial examination (50%)<br><br>or<br><br>Resit = integral exam (100%)<br><br>The case study assignment is graded with a pass/fail.<br><br>minimum single partial examination score must be 5.0<br><br>For more information on grading, see article 14 in the Rules and Guidelines (RGBE):<br><a href="https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations">https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations</a> |
| <b>Judgement</b>                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Permitted Materials during Exam</b>  | Classical Calculators (No tablet computers or mobile devices are allowed)<br>Aid sheet: An aid sheet (Exam version) will be provided by the lecturers on Brightspace. Students are allowed to print this aid sheet to use it during the examination. Therefore no self-made aid sheet is allowed                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Collegerama</b>                      | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

| <b>CTB1320-24 Toets 1</b>               | <b>Partial exam 1</b>                | <b>0</b> |
|-----------------------------------------|--------------------------------------|----------|
| <b>Responsible Instructor</b>           | Dr. O. Copuroglu                     |          |
| <b>Contact Hours / Week</b><br>x/x/x/x  | Check CTB1320-17                     |          |
| <b>Education Period</b>                 | 3                                    |          |
| <b>Start Education</b>                  | 3                                    |          |
| <b>Exam Period</b>                      | 3                                    |          |
| <b>Course Language</b>                  | Dutch<br>English                     |          |
| <b>Course Contents</b>                  | Check CTB1320-17                     |          |
| <b>Study Goals</b>                      | Check CTB1320-17                     |          |
| <b>Education Method</b>                 | Check CTB1320-17                     |          |
| <b>Assessment</b>                       | TOETS 01 - deoltoets 01 - WEIGHT=50% |          |
| <b>Expected prior Knowledge</b>         | Check CTB1320-17                     |          |
| <b>Academic Skills</b>                  | Check CTB1320-17                     |          |
| <b>Literature &amp; Study Materials</b> | Check CTB1320-17                     |          |
| <b>Judgement</b>                        | Check CTB1320-17                     |          |
| <b>Permitted Materials during Exam</b>  | Check CTB1320-17                     |          |
| <b>Collegerama</b>                      | No                                   |          |

| <b>CTB1320-24 Toets 2</b>               | <b>Partial exam 2</b>                | <b>0</b> |
|-----------------------------------------|--------------------------------------|----------|
| <b>Responsible Instructor</b>           | Dr. O. Copuroglu                     |          |
| <b>Contact Hours / Week</b><br>x/x/x/x  | Check CTB1320-17                     |          |
| <b>Education Period</b>                 | 3                                    |          |
| <b>Start Education</b>                  | 3                                    |          |
| <b>Exam Period</b>                      | 3                                    |          |
| <b>Course Language</b>                  | Dutch<br>English                     |          |
| <b>Course Contents</b>                  | Check CTB1320-17                     |          |
| <b>Study Goals</b>                      | Check CTB1320-17                     |          |
| <b>Education Method</b>                 | Check CTB1320-17                     |          |
| <b>Assessment</b>                       | TOETS 02 - deoltoets 02 - WEIGHT=50% |          |
| <b>Expected prior Knowledge</b>         | Check CTB1320-17                     |          |
| <b>Academic Skills</b>                  | Check CTB1320-17                     |          |
| <b>Literature &amp; Study Materials</b> | Check CTB1320-17                     |          |
| <b>Judgement</b>                        | Check CTB1320-17                     |          |
| <b>Permitted Materials during Exam</b>  | Check CTB1320-17                     |          |
| <b>Collegerama</b>                      | No                                   |          |

| <b>CTB1320-24 Toets 3</b>               | <b>Ethics</b>                         | <b>0</b> |
|-----------------------------------------|---------------------------------------|----------|
| <b>Responsible Instructor</b>           | Dr. O. Copuroglu                      |          |
| <b>Contact Hours / Week</b><br>x/x/x/x  | Check CTB1320-17                      |          |
| <b>Education Period</b>                 | 3                                     |          |
| <b>Start Education</b>                  | 3                                     |          |
| <b>Exam Period</b>                      | 3                                     |          |
| <b>Course Language</b>                  | Dutch<br>English                      |          |
| <b>Course Contents</b>                  | Check CTB1320-17                      |          |
| <b>Study Goals</b>                      | Check CTB1320-17                      |          |
| <b>Education Method</b>                 | Check CTB1320-17                      |          |
| <b>Assessment</b>                       | TOETS 03 ETHICS WEIGHT=0% (PASS/FAIL) |          |
| <b>Expected prior Knowledge</b>         | Check CTB1320-17                      |          |
| <b>Academic Skills</b>                  | Check CTB1320-17                      |          |
| <b>Literature &amp; Study Materials</b> | Check CTB1320-17                      |          |
| <b>Judgement</b>                        | Check CTB1320-17                      |          |
| <b>Permitted Materials during Exam</b>  | Check CTB1320-17                      |          |
| <b>Collegerama</b>                      | No                                    |          |

| <b>CTB1320-24 Toets 4</b>               | <b>Case study</b>                                  | <b>0</b> |
|-----------------------------------------|----------------------------------------------------|----------|
| <b>Responsible Instructor</b>           | Dr. O. Copuroglu                                   |          |
| <b>Contact Hours / Week</b><br>x/x/x/x  | Check CTB1320-17                                   |          |
| <b>Education Period</b>                 | 3                                                  |          |
| <b>Start Education</b>                  | 3                                                  |          |
| <b>Exam Period</b>                      | 3                                                  |          |
| <b>Course Language</b>                  | Dutch<br>English                                   |          |
| <b>Course Contents</b>                  | Check CTB1320-17                                   |          |
| <b>Study Goals</b>                      | Check CTB1320-17                                   |          |
| <b>Education Method</b>                 | Check CTB1320-17                                   |          |
| <b>Assessment</b>                       | TOETS 04 - CASE STUDY - WEIGHT=0% (PASS/FAIL). 0EC |          |
| <b>Expected prior Knowledge</b>         | Check CTB1320-17                                   |          |
| <b>Academic Skills</b>                  | Check CTB1320-17                                   |          |
| <b>Literature &amp; Study Materials</b> | Check CTB1320-17                                   |          |
| <b>Judgement</b>                        | Check CTB1320-17                                   |          |
| <b>Permitted Materials during Exam</b>  | Check CTB1320-17                                   |          |
| <b>Collegerama</b>                      | No                                                 |          |

| <b>CTB1320-24 Toets 5</b>               | <b>Integral exam</b>                    | <b>5</b> |
|-----------------------------------------|-----------------------------------------|----------|
| <b>Responsible Instructor</b>           | Dr. O. Copuroglu                        |          |
| <b>Contact Hours / Week</b><br>x/x/x/x  | Check CTB1320-17                        |          |
| <b>Education Period</b>                 | 3                                       |          |
| <b>Start Education</b>                  | 3                                       |          |
| <b>Exam Period</b>                      | 3                                       |          |
| <b>Course Language</b>                  | Dutch<br>English                        |          |
| <b>Course Contents</b>                  | Check CTB1320-17                        |          |
| <b>Study Goals</b>                      | Check CTB1320-17                        |          |
| <b>Education Method</b>                 | Check CTB1320-17                        |          |
| <b>Assessment</b>                       | TOETS 05 - INTEGRAL - WEIGHT=100% (5EC) |          |
| <b>Expected prior Knowledge</b>         | Check CTB1320-17                        |          |
| <b>Academic Skills</b>                  | Check CTB1320-17                        |          |
| <b>Literature &amp; Study Materials</b> | Check CTB1320-17                        |          |
| <b>Judgement</b>                        | Check CTB1320-17                        |          |
| <b>Permitted Materials during Exam</b>  | Check CTB1320-17                        |          |
| <b>Collegerama</b>                      | No                                      |          |

| <b>CTB1410-20</b>             | <b>Design of Structures and Foundations 1</b> | <b>5</b> |
|-------------------------------|-----------------------------------------------|----------|
| <b>Responsible Instructor</b> | Dr.ir. H.R. Schipper                          |          |
| <b>Instructor</b>             | Drs.ir. R.E.P. de Nijs                        |          |
| <b>Instructor</b>             | L. Flessati                                   |          |
| <b>Education Period</b>       | 4                                             |          |
| <b>Start Education</b>        | 4                                             |          |
| <b>Exam Period</b>            | 4<br>5                                        |          |
| <b>Course Language</b>        | Dutch                                         |          |
| <b>Collegerama</b>            | Yes                                           |          |

| <b>CTB1410-20 Toets 1</b>     | <b>Constructies en Funderingen</b> | <b>3</b> |
|-------------------------------|------------------------------------|----------|
| <b>Responsible Instructor</b> | Dr.ir. H.R. Schipper               |          |
| <b>Education Period</b>       | 4                                  |          |
| <b>Start Education</b>        | 4                                  |          |
| <b>Exam Period</b>            | 4<br>5                             |          |
| <b>Course Language</b>        | Dutch                              |          |
| <b>Collegerama</b>            | No                                 |          |

| <b>CTB1410-20 Toets 2</b>     | <b>Ontwerppracticum</b> | <b>2</b> |
|-------------------------------|-------------------------|----------|
| <b>Responsible Instructor</b> | Dr.ir. H.R. Schipper    |          |
| <b>Education Period</b>       | 4                       |          |
| <b>Start Education</b>        | 4                       |          |
| <b>Exam Period</b>            | 4<br>5                  |          |
| <b>Course Language</b>        | Dutch                   |          |
| <b>Collegerama</b>            | No                      |          |

| <b>CTB1420-17</b>             | <b>Transport en Planning</b> | <b>5</b> |
|-------------------------------|------------------------------|----------|
| <b>Responsible Instructor</b> | Dr. V.L. Knoop               |          |
| <b>Instructor</b>             | Dr.ir. N. van Oort           |          |
| <b>Practical Coordinator</b>  | Dr.ir. A. Gavrilidou         |          |
| <b>Education Period</b>       | 4                            |          |
| <b>Start Education</b>        | 4                            |          |
| <b>Exam Period</b>            | 4                            |          |
| <b>Course Language</b>        | Dutch                        |          |
| <b>Collegerama</b>            | No                           |          |



| <b>CTB1420-17 Toets 1</b> | <b><i>Deeltentamen Delen 1 en 2 van het dictaat</i></b> | <b>0</b> |
|---------------------------|---------------------------------------------------------|----------|
| Responsible Instructor    | Dr. V.L. Knoop                                          |          |
| Education Period          | 4                                                       |          |
| Start Education           | 4                                                       |          |
| Exam Period               | 4                                                       |          |
| Course Language           | Dutch                                                   |          |
| Collegerama               | No                                                      |          |

| <b>CTB1420-17 Toets 2</b> | <b><i>Deeltentamen Delen 3 en 4 van het dictaat</i></b> | <b>0</b> |
|---------------------------|---------------------------------------------------------|----------|
| Responsible Instructor    | Dr. V.L. Knoop                                          |          |
| Education Period          | 4                                                       |          |
| Start Education           | 4                                                       |          |
| Exam Period               | 4                                                       |          |
| Course Language           | Dutch                                                   |          |
| Collegerama               | No                                                      |          |

| <b>CTB1420-17 Toets 3</b> | <b><i>Ethics CTB1420-17</i></b> | <b>0</b> |
|---------------------------|---------------------------------|----------|
| Responsible Instructor    | Dr. V.L. Knoop                  |          |
| Education Period          | 4                               |          |
| Start Education           | 4                               |          |
| Exam Period               | 4                               |          |
| Course Language           | Dutch                           |          |
| Collegerama               | No                              |          |

| <b>CTB1420-17 Toets 4</b> | <b><i>Oefening CTB1420-17</i></b> | <b>1</b> |
|---------------------------|-----------------------------------|----------|
| Responsible Instructor    | Dr. V.L. Knoop                    |          |
| Education Period          | 4                                 |          |
| Start Education           | 4                                 |          |
| Exam Period               | 4                                 |          |
| Course Language           | Dutch                             |          |
| Collegerama               | No                                |          |

| <b>CTB1420-17 Toets 5</b> | <b><i>Integrale Toets Delen 1 t/m 5</i></b> | <b>4</b> |
|---------------------------|---------------------------------------------|----------|
| Responsible Instructor    | Dr. V.L. Knoop                              |          |
| Education Period          | 4                                           |          |
| Start Education           | 4                                           |          |
| Exam Period               | 4                                           |          |
| Course Language           | Dutch                                       |          |
| Collegerama               | No                                          |          |

|                     |                                          |
|---------------------|------------------------------------------|
| <b>Year</b>         | <b>2024/2025</b>                         |
| <b>Organization</b> | <b>Civil Engineering and Geosciences</b> |
| <b>Education</b>    | <b>Bachelor Civil Engineering</b>        |

**CT BSc 2e jaar 2024**

| <b>CTB2000-20</b>             | <b>Construction Site 2</b> | <b>8</b> |
|-------------------------------|----------------------------|----------|
| <b>Responsible Instructor</b> | Dr. V.L. Markine           |          |
| <b>Instructor</b>             | Dr. S.L.M. Lhermitte       |          |
| <b>Instructor</b>             | Dr.ir. T.S. van den Bremer |          |
| <b>Instructor</b>             | Ir. P.A. de Vries          |          |
| <b>Instructor</b>             | Prof.dr. H.M. Jonkers      |          |
| <b>Instructor</b>             | Prof.dr. P.J. Vardon       |          |
| <b>Education Period</b>       | 1<br>2<br>3<br>4           |          |
| <b>Start Education</b>        | 1<br>2<br>3<br>4           |          |
| <b>Exam Period</b>            | none                       |          |
| <b>Course Language</b>        | Dutch                      |          |
| <b>Collegerama</b>            | No                         |          |

| <b>CTB2000-20-D1a</b>         | <b>Programmeren 1</b> | <b>1</b> |
|-------------------------------|-----------------------|----------|
| <b>Responsible Instructor</b> | Dr. V.L. Markine      |          |
| <b>Education Period</b>       | 1                     |          |
| <b>Start Education</b>        | 1                     |          |
| <b>Exam Period</b>            | none                  |          |
| <b>Course Language</b>        | Dutch                 |          |
| <b>Collegerama</b>            | No                    |          |

| <b>CTB2000-20-D1b</b>         | <b>Vloeistofmechanica practicum</b> | <b>1</b> |
|-------------------------------|-------------------------------------|----------|
| <b>Responsible Instructor</b> | Dr. V.L. Markine                    |          |
| <b>Instructor</b>             | Dr.ir. T.S. van den Bremer          |          |
| <b>Education Period</b>       | 1                                   |          |
| <b>Start Education</b>        | 1                                   |          |
| <b>Exam Period</b>            | none                                |          |
| <b>Course Language</b>        | Dutch                               |          |
| <b>Collegerama</b>            | No                                  |          |

| <b>CTB2000-20-D2a</b>         | <b>Programmeren 2</b> | <b>1</b> |
|-------------------------------|-----------------------|----------|
| <b>Responsible Instructor</b> | Dr. V.L. Markine      |          |
| <b>Education Period</b>       | 2                     |          |
| <b>Start Education</b>        | 2                     |          |
| <b>Exam Period</b>            | none                  |          |
| <b>Course Language</b>        | Dutch                 |          |
| <b>Collegerama</b>            | No                    |          |

| <b>CTB2000-20-D2b</b>         | <b>Materiaalkunde practicum</b> | <b>1</b> |
|-------------------------------|---------------------------------|----------|
| <b>Responsible Instructor</b> | Dr. V.L. Markine                |          |
| <b>Instructor</b>             | Ir. P.A. de Vries               |          |
| <b>Instructor</b>             | Prof.dr. H.M. Jonkers           |          |
| <b>Education Period</b>       | 2                               |          |
| <b>Start Education</b>        | 2                               |          |
| <b>Exam Period</b>            | none                            |          |
| <b>Course Language</b>        | Dutch                           |          |
| <b>Collegerama</b>            | No                              |          |

| <b>CTB2000-20-D3a</b>         | <b>Programmeren 3</b> | <b>1</b> |
|-------------------------------|-----------------------|----------|
| <b>Responsible Instructor</b> | Dr. V.L. Markine      |          |
| <b>Education Period</b>       | 3                     |          |
| <b>Start Education</b>        | 3                     |          |
| <b>Exam Period</b>            | none                  |          |
| <b>Course Language</b>        | Dutch                 |          |
| <b>Collegerama</b>            | No                    |          |

| <b>CTB2000-20-D3b</b>         | <b><i>Grondmechanica practicum</i></b> | <b>1</b> |
|-------------------------------|----------------------------------------|----------|
| <b>Responsible Instructor</b> | Dr. V.L. Markine                       |          |
| <b>Instructor</b>             | Prof.dr. P.J. Vardon                   |          |
| <b>Education Period</b>       | 3                                      |          |
| <b>Start Education</b>        | 3                                      |          |
| <b>Exam Period</b>            | none                                   |          |
| <b>Course Language</b>        | Dutch                                  |          |
| <b>Collegerama</b>            | No                                     |          |

| <b>CTB2000-20-D4a</b>         | <b><i>Programmeren 4</i></b> | <b>1</b> |
|-------------------------------|------------------------------|----------|
| <b>Responsible Instructor</b> | Dr. V.L. Markine             |          |
| <b>Instructor</b>             | Dr. rer. nat. G. George      |          |
| <b>Education Period</b>       | 4                            |          |
| <b>Start Education</b>        | 4                            |          |
| <b>Exam Period</b>            | none                         |          |
| <b>Course Language</b>        | Dutch                        |          |
| <b>Collegerama</b>            | No                           |          |

| <b>CTB2000-20-D4b</b>         | <b><i>Numerieke wiskunde practicum</i></b> | <b>1</b> |
|-------------------------------|--------------------------------------------|----------|
| <b>Responsible Instructor</b> | Dr. V.L. Markine                           |          |
| <b>Instructor</b>             | Dr. O. Jaibi                               |          |
| <b>Education Period</b>       | 4                                          |          |
| <b>Start Education</b>        | 4                                          |          |
| <b>Exam Period</b>            | none                                       |          |
| <b>Course Language</b>        | Dutch                                      |          |
| <b>Collegerama</b>            | No                                         |          |

| <b>CTB2105</b>                         | <b><i>Differentiaalvergelijkingen</i></b> | <b>3</b> |
|----------------------------------------|-------------------------------------------|----------|
| <b>Responsible Instructor</b>          | Dr. O. Jaibi                              |          |
| <b>Co-responsible for assignments</b>  | Dr.ir. M. Keijzer                         |          |
| <b>Contact Hours / Week</b><br>x/x/x/x | 4/0/0/0                                   |          |
| <b>Education Period</b>                | 1                                         |          |
| <b>Start Education</b>                 | 1                                         |          |
| <b>Exam Period</b>                     | 1<br>2                                    |          |
| <b>Course Language</b>                 | Dutch                                     |          |
| <b>Collegerama</b>                     | No                                        |          |

| <b>CTB2105 Toets 1</b>                | <b><i>Exam</i></b> | <b>3</b> |
|---------------------------------------|--------------------|----------|
| <b>Responsible Instructor</b>         | Dr. O. Jaibi       |          |
| <b>Co-responsible for assignments</b> | Dr.ir. M. Keijzer  |          |
| <b>Education Period</b>               | 1                  |          |
| <b>Start Education</b>                | 1                  |          |
| <b>Exam Period</b>                    | 1<br>2             |          |
| <b>Course Language</b>                | Dutch              |          |
| <b>Collegerama</b>                    | No                 |          |

| <b>CTB2105 Toets 2</b>                | <b><i>Practical</i></b> | <b>0</b> |
|---------------------------------------|-------------------------|----------|
| <b>Responsible Instructor</b>         | Dr. O. Jaibi            |          |
| <b>Co-responsible for assignments</b> | Dr.ir. M. Keijzer       |          |
| <b>Education Period</b>               | 1                       |          |
| <b>Start Education</b>                | 1                       |          |
| <b>Exam Period</b>                    | 1                       |          |
| <b>Course Language</b>                | Dutch                   |          |
| <b>Collegerama</b>                    | No                      |          |

|                               |                            |          |
|-------------------------------|----------------------------|----------|
| <b>CTB2110</b>                | <b>Fluid Mechanics</b>     | <b>5</b> |
| <b>Responsible Instructor</b> | Dr. C. Chassagne           |          |
| <b>Responsible Instructor</b> | Dr.ir. T.S. van den Bremer |          |
| <b>Instructor</b>             | Dr.ir. T.S. van den Bremer |          |
| <b>Instructor</b>             | Dr. C. Chassagne           |          |
| <b>Education Period</b>       | 1                          |          |
| <b>Start Education</b>        | 1                          |          |
| <b>Exam Period</b>            | 1<br>2                     |          |
| <b>Course Language</b>        | Dutch                      |          |
| <b>Collegerama</b>            | Yes                        |          |

|                               |                                                 |          |
|-------------------------------|-------------------------------------------------|----------|
| <b>CTB2121</b>                | <b>Urban Water and Environmental Technology</b> | <b>5</b> |
| <b>Responsible Instructor</b> | Prof.dr.ir. L.C. Rietveld                       |          |
| <b>Instructor</b>             | Prof.dr.ir. M.K. de Kreuk                       |          |
| <b>Instructor</b>             | Prof.dr.ir. D. van Halem                        |          |
| <b>Instructor</b>             | Prof.dr.ir. J.B. van Lier                       |          |
| <b>Education Period</b>       | 1                                               |          |
| <b>Start Education</b>        | 1                                               |          |
| <b>Exam Period</b>            | 1                                               |          |
| <b>Course Language</b>        | Dutch                                           |          |
| <b>Collegerama</b>            | No                                              |          |

|                               |                                     |          |
|-------------------------------|-------------------------------------|----------|
| <b>CTB2121 Toets 1</b>        | <b>Deeltentamen Stedelijk Water</b> | <b>0</b> |
| <b>Responsible Instructor</b> | Prof.dr.ir. L.C. Rietveld           |          |
| <b>Instructor</b>             | Prof.dr.ir. M.K. de Kreuk           |          |
| <b>Instructor</b>             | Prof.dr.ir. D. van Halem            |          |
| <b>Instructor</b>             | Prof.dr.ir. J.B. van Lier           |          |
| <b>Education Period</b>       | 1                                   |          |
| <b>Start Education</b>        | 1                                   |          |
| <b>Exam Period</b>            | 1                                   |          |
| <b>Course Language</b>        | Dutch                               |          |
| <b>Collegerama</b>            | No                                  |          |

|                               |                                    |          |
|-------------------------------|------------------------------------|----------|
| <b>CTB2121 Toets 2</b>        | <b>Deeltentamen Milieutechniek</b> | <b>0</b> |
| <b>Responsible Instructor</b> | Prof.dr.ir. L.C. Rietveld          |          |
| <b>Instructor</b>             | Prof.dr.ir. M.K. de Kreuk          |          |
| <b>Instructor</b>             | Prof.dr.ir. D. van Halem           |          |
| <b>Instructor</b>             | Prof.dr.ir. J.B. van Lier          |          |
| <b>Education Period</b>       | 1                                  |          |
| <b>Start Education</b>        | 1                                  |          |
| <b>Exam Period</b>            | 1                                  |          |
| <b>Course Language</b>        | Dutch                              |          |
| <b>Collegerama</b>            | No                                 |          |

|                               |                           |          |
|-------------------------------|---------------------------|----------|
| <b>CTB2121 Toets 3</b>        | <b>Integraal tentamen</b> | <b>3</b> |
| <b>Responsible Instructor</b> | Prof.dr.ir. L.C. Rietveld |          |
| <b>Instructor</b>             | Prof.dr.ir. M.K. de Kreuk |          |
| <b>Instructor</b>             | Prof.dr.ir. D. van Halem  |          |
| <b>Instructor</b>             | Prof.dr.ir. J.B. van Lier |          |
| <b>Education Period</b>       | 1                         |          |
| <b>Start Education</b>        | 1                         |          |
| <b>Exam Period</b>            | 1                         |          |
| <b>Course Language</b>        | Dutch                     |          |
| <b>Collegerama</b>            | No                        |          |

| <b>CTB2121 Toets 4</b>        | <b>Integrale opdracht</b> | <b>2</b> |
|-------------------------------|---------------------------|----------|
| <b>Responsible Instructor</b> | Prof.dr.ir. L.C. Rietveld |          |
| <b>Education Period</b>       | 1                         |          |
| <b>Start Education</b>        | 1                         |          |
| <b>Exam Period</b>            | 1                         |          |
| <b>Course Language</b>        | Dutch                     |          |
| <b>Collegerama</b>            | No                        |          |

| <b>CTB2121 Toets 5</b>        | <b>Workshop</b>           | <b>0</b> |
|-------------------------------|---------------------------|----------|
| <b>Responsible Instructor</b> | Prof.dr.ir. L.C. Rietveld |          |
| <b>Education Period</b>       | 1                         |          |
| <b>Start Education</b>        | 1                         |          |
| <b>Exam Period</b>            | 1                         |          |
| <b>Course Language</b>        | Dutch                     |          |
| <b>Collegerama</b>            | No                        |          |

| <b>CTB2200</b>                        | <b>Kansrekening &amp; Statistiek</b> | <b>3</b> |
|---------------------------------------|--------------------------------------|----------|
| <b>Responsible Instructor</b>         | Dr. C.O. Smet                        |          |
| <b>Co-responsible for assignments</b> | Dr.ir. N.P.K. Chan                   |          |
| <b>Education Period</b>               | 2                                    |          |
| <b>Start Education</b>                | 2                                    |          |
| <b>Exam Period</b>                    | 2<br>3                               |          |
| <b>Course Language</b>                | Dutch                                |          |
| <b>Collegerama</b>                    | No                                   |          |

| <b>CTB2210</b>                | <b>Construction Mechanics 3</b> | <b>5</b> |
|-------------------------------|---------------------------------|----------|
| <b>Responsible Instructor</b> | Prof.dr.ir. J.G. Rots           |          |
| <b>Education Period</b>       | 2                               |          |
| <b>Start Education</b>        | 2                               |          |
| <b>Exam Period</b>            | 2<br>3                          |          |
| <b>Course Language</b>        | Dutch                           |          |
| <b>Collegerama</b>            | Yes                             |          |

| <b>CTB2220-14</b>             | <b>Concrete and Steel Structures</b> | <b>5</b> |
|-------------------------------|--------------------------------------|----------|
| <b>Responsible Instructor</b> | Dr.ir. M.J.M.M. Steenbergen          |          |
| <b>Instructor</b>             | Dr. F. Kavoura                       |          |
| <b>Instructor</b>             | Ir. P.A. de Vries                    |          |
| <b>Instructor</b>             | Prof.dr.ir. M.A.N. Hendriks          |          |
| <b>Instructor</b>             | Dr. M. Pavlovic                      |          |
| <b>Education Period</b>       | 2                                    |          |
| <b>Start Education</b>        | 2                                    |          |
| <b>Exam Period</b>            | 2                                    |          |
| <b>Course Language</b>        | Dutch<br>English                     |          |
| <b>Collegerama</b>            | Yes                                  |          |

| <b>CTB2220-14 Toets 1</b>     | <b>Deeltoets 1</b>          | <b>0</b> |
|-------------------------------|-----------------------------|----------|
| <b>Responsible Instructor</b> | Dr.ir. M.J.M.M. Steenbergen |          |
| <b>Instructor</b>             | Dr. M. Pavlovic             |          |
| <b>Instructor</b>             | Dr. F. Kavoura              |          |
| <b>Instructor</b>             | Ir. P.A. de Vries           |          |
| <b>Instructor</b>             | Prof.dr.ir. M.A.N. Hendriks |          |
| <b>Education Period</b>       | 2                           |          |
| <b>Start Education</b>        | 2                           |          |
| <b>Exam Period</b>            | 2                           |          |
| <b>Course Language</b>        | Dutch                       |          |
| <b>Collegerama</b>            | No                          |          |

| <b>CTB2220-14 Toets 2</b>     | <b>Deeltoets 2</b>          | <b>0</b> |
|-------------------------------|-----------------------------|----------|
| <b>Responsible Instructor</b> | Dr.ir. M.J.M.M. Steenbergen |          |
| <b>Instructor</b>             | Dr. M. Pavlovic             |          |
| <b>Instructor</b>             | Dr. F. Kavoura              |          |
| <b>Instructor</b>             | Ir. P.A. de Vries           |          |
| <b>Instructor</b>             | Prof.dr.ir. M.A.N. Hendriks |          |
| <b>Education Period</b>       | 2                           |          |
| <b>Start Education</b>        | 2                           |          |
| <b>Exam Period</b>            | 2                           |          |
| <b>Course Language</b>        | Dutch                       |          |
| <b>Collegerama</b>            | No                          |          |

| <b>CTB2220-14 Toets 3</b>     | <b>Oefening</b>             | <b>0</b> |
|-------------------------------|-----------------------------|----------|
| <b>Responsible Instructor</b> | Dr.ir. M.J.M.M. Steenbergen |          |
| <b>Instructor</b>             | Dr. M. Pavlovic             |          |
| <b>Instructor</b>             | Dr. F. Kavoura              |          |
| <b>Instructor</b>             | Ir. P.A. de Vries           |          |
| <b>Instructor</b>             | Prof.dr.ir. M.A.N. Hendriks |          |
| <b>Education Period</b>       | 2                           |          |
| <b>Start Education</b>        | 2                           |          |
| <b>Exam Period</b>            | 2                           |          |
| <b>Course Language</b>        | Dutch                       |          |
| <b>Collegerama</b>            | No                          |          |

| <b>CTB2220-14 Toets 4</b>     | <b>Integrale Toets</b>      | <b>5</b> |
|-------------------------------|-----------------------------|----------|
| <b>Responsible Instructor</b> | Dr.ir. M.J.M.M. Steenbergen |          |
| <b>Instructor</b>             | Dr. M. Pavlovic             |          |
| <b>Instructor</b>             | Dr. F. Kavoura              |          |
| <b>Instructor</b>             | Ir. P.A. de Vries           |          |
| <b>Instructor</b>             | Prof.dr.ir. M.A.N. Hendriks |          |
| <b>Education Period</b>       | 2                           |          |
| <b>Start Education</b>        | 2                           |          |
| <b>Exam Period</b>            | 2                           |          |
| <b>Course Language</b>        | Dutch                       |          |
| <b>Collegerama</b>            | No                          |          |

| CTB2300                                 | Dynamics of Systems                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 3 |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>           | Dr.ir. K.N. van Dalen                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Responsible Instructor</b>           | Dr.ir. H. Hendrikse                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Instructor</b>                       | Dr.ir. H. Hendrikse                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Co-responsible for assignments</b>   | Dr.ir. H. Hendrikse                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Contact Hours / Week</b><br>x/x/x/x  | 0/0/4/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Education Period</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Start Education</b>                  | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Exam Period</b>                      | 3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Course Language</b>                  | Dutch<br>English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Expected prior knowledge</b>         | Dynamics and Modelling<br>Differential Equations for civil engineering                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Parts</b>                            | The course consists of lectures in which the course content is provided.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Summary</b>                          | To be able to model and analyse the dynamic behaviour of several civil engineering systems                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Course Contents</b>                  | Mechanical system with one degree of freedom, undamped. Formulate equations of motion for free vibration. Forced vibrations for harmonic, exponential, step and impact/pulse loads. Application of initial conditions. Hydraulic systems with one degree of freedom without damping; also free and forced motion.<br>Mechanical system with damping; free and forced vibrations.<br>Hydraulic system with damping; free and forced motion.<br>First order systems as a limit case of 2nd-order systems.<br>Mechanical systems with two degrees of freedom, without damping.<br>Formulate equations of motion (mass matrix, stiffness matrix).<br>Determination of eigenfrequencies and eigenperiods. Forced response for harmonic loads.                                                                                                                  |   |
| <b>Study Goals</b>                      | Learning Objectives (LO):<br><br>LO1: To recognize the properties of structural and hydraulic civil engineering systems (for example mass, elasticity, discharges), that are essential for the modelling of such systems.<br>LO2: To be able to formulate simple equations of motion, using the second Newtons Law, with the help of the displacement method.<br>LO3: To be able to analytically derive solutions for formulated equations of motion.<br>LO4: Interpret solutions with respect to the fundamental dynamics parameters, like: eigenfrequency, eigenperiod, free and forced motion, resonance, and steady-state motion.<br>LO5: To be able to predict design improvements for dynamic systems that have been studied.<br>30 hours lectures<br>3 hours exam<br>51 hours individual study and exam preparation<br>-----+<br>84 hours (3 ECTS) |   |
| <b>Education Method</b>                 | Lectures, tutorials                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Course Relations</b>                 | Dynamics<br>Structural mechanics courses                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Literature and Study Materials</b>   | Study guide with general information (BrightSpace)<br>Lecture Notes: "Dynamica van Systemen", J. Blaauwendraad, versie 2010.<br>Lecture Slides<br>Other course materials:<br>"Dynamica van Systemen, Vragenstukkenbundel (oefen- en tentamenopgaven)", Studentassistenten CT2022, versie 2011.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Assessment</b>                       | Written exam, with open questions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Permitted Materials during Tests</b> | Calculator and a sheet with main formulas. The latter will be given at the exam together with the exam questions.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Expected prior Knowledge</b>         | Dynamica & Modelforming; Constructuemechanica 1,2,3; Differentiaalvergelijkingen.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Academic Skills</b>                  | Analytical thinking; Critical appraisal.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Literature &amp; Study Materials</b> | Study guide with general information (BrightSpace)<br>Lecture Notes: "Dynamica van Systemen", J. Blaauwendraad, versie 2010.<br>Lecture Slides<br>Other course materials:<br>"Dynamica van Systemen, Vragenstukkenbundel (oefen- en tentamenopgaven)", Studentassistenten CT2022, versie 2011.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Judgement</b>                        | For more information on grading, see article 14 in the Rules and Guidelines (RGBE):<br><a href="https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations">https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Permitted Materials during Exam</b>  | Calculator and a sheet with main formulas. The latter will be given at the exam together with the exam questions.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Colleggerama</b>                     | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |



| CTB2310                                 | Soil Mechanics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 5 |
|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>           | Prof.dr. M.A. Hicks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Instructor</b>                       | Prof.dr. P.J. Vardon                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Contact Hours / Week</b><br>x/x/x/x  | 0/0/6/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Education Period</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Start Education</b>                  | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Exam Period</b>                      | 3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Course Language</b>                  | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Parts</b>                            | The course is split into 2 parts: the first part is largely equivalent to the old Soil Mechanics 1 course; the second part is largely equivalent to the old Soil Mechanics 2 course.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Summary</b>                          | The course explains the basic concepts of theoretical and applied soil mechanics. Theoretical soil mechanics includes: Soil Characteristics; Groundwater; Geomechanics; Shear Strength of Soils. Applied soil mechanics includes: Retaining Structures; Foundations; Slope Stability.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Course Contents</b>                  | Soil Characteristics: Classification and mineralogy of soils; Grains, grain size distribution and porosity; Introduction to in situ testing.<br><br>Groundwater: Pore pressure and effective stress; Darcys law, permeability and groundwater flow.<br><br>Geomechanics: Stresses and initial stress state; Strains, stress-strain relationships and tangent modulus; Elastic solutions; Consolidation, drained and undrained behaviour.<br><br>Shear Strength of Soils: Drained and undrained soil behaviour; total and effective shear strength parameters; Mohr circles; Mohr-Coulomb failure criterion; stress paths; shear test; triaxial test; site investigation; soil sampling.<br><br>Retaining Structures: Lateral earth pressure at rest; passive and active pressures; Rankine; Coulomb; sheet pile walls; Blum.<br><br>Foundations: Geotechnical bearing capacity; shallow and deep foundations; Prandtl; Brinch Hansen; pile foundations.<br><br>Slope Stability: Limit equilibrium methods; vertical slope; infinite slope; method of slices; Fellenius; Bishop.                                                                                                                   |   |
| <b>Study Goals</b>                      | The main goal of this course is to:<br>Have an understanding of, and be able to apply, the following: Soil classification; In situ testing; Groundwater flow; Stresses in the ground; Strains and stiffness; Consolidation theory; Shear strength of soils; Laboratory and in situ testing; Earth retaining structures; Foundations; Slope stability.<br><br>By the end of the course the student should be able to:<br>Explain the various forms of soil behaviour, including the influence of soil properties, degree of saturation and loading rate.<br>Evaluate laboratory data to determine material parameter values.<br>Plan an engineering strategy for analysing geotechnical problems, involving settlements, consolidation, groundwater flow, basic foundations, retaining structures and slope stability.<br>Analyse appropriately and evaluate geotechnical problems, utilising presented theory, on the following topics: elastic behaviour and settlements, consolidation, groundwater flow, basic foundations, retaining structures and slope stability.<br><br>Study load:<br>36 lecture hours, 12 tutorial hours, 72 self-study hours, 17 exam preparation hours, 3 exam hours. |   |
| <b>Education Method</b>                 | Lectures, tutorials and self-study.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Course Relations</b>                 | Design of Structures and Foundations 1 & 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Assessment</b>                       | In 'de Bouwplaats' a soil mechanics practical will be executed<br><br>The learning goals of this course will be tested with:<br>Written examination (in English/Dutch) at the end of the course.<br>Mid-term test must be passed to gain entry to the exam.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Permitted Materials during Tests</b> | Calculator as described in the Examination regulations<br>Writing and drawing implements.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Expected prior Knowledge</b>         | Structural Mechanics 1<br>Structural Mechanics 2<br>Fluid Mechanics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Academic Skills</b>                  | Analytical skills.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Literature &amp; Study Materials</b> | Course book (optional): Soil Mechanics, A. Verruijt, Delft University of Technology, 2012. Go to: <a href="http://geo.verruijt.net/">http://geo.verruijt.net/</a> and download the book (in English) stored as SoilMechBook2012.pdf<br>Accompanying lectures notes (in the form of ppt slides).<br>Worked examples.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Judgement</b>                        | Mid-term test must be passed to gain entry to the exam. Final mark = examination grade only.<br><br>For more information on grading, see article 14 in the Rules and Guidelines (RGE):<br><a href="https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations">https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Permitted Materials during Exam</b>  | Standard calculator. Formula sheet will be provided.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Colleggerama</b>                     | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |

| CTB2320-17                      | Design of Structures and Foundations 2                                                                                                                                      | 5 |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| Responsible Instructor          | R. Oval                                                                                                                                                                     |   |
| Instructor                      | Dr.ir. M. Korff                                                                                                                                                             |   |
| Instructor                      | Dr.ing. K.C. Terwel                                                                                                                                                         |   |
| Instructor                      | Dr.ir. H.R. Schipper                                                                                                                                                        |   |
| Instructor                      | H. Alkisaai                                                                                                                                                                 |   |
| Instructor                      | Prof.dr. K.G. Gavin                                                                                                                                                         |   |
| Contact Hours / Week<br>x/x/x/x | 0/0/10/0                                                                                                                                                                    |   |
| Education Period                | 3                                                                                                                                                                           |   |
| Start Education                 | 3                                                                                                                                                                           |   |
| Exam Period                     | 3                                                                                                                                                                           |   |
| Course Language                 | Dutch                                                                                                                                                                       |   |
| Tags                            | Building Engineering<br>Challenging<br>Design<br>Drawing<br>Geo Engineering<br>Intensive<br>Practicals<br>Projects<br>Structural Mechanics<br>Sustainability<br>Underground |   |
| Collegerama                     | Yes                                                                                                                                                                         |   |

| CTB2320-17 Toets 1     | Deeltoets 1 | 0 |
|------------------------|-------------|---|
| Responsible Instructor | R. Oval     |   |
| Education Period       | 3           |   |
| Start Education        | 3           |   |
| Exam Period            | 3           |   |
| Course Language        | Dutch       |   |
| Collegerama            | Yes         |   |

| CTB2320-17 Toets 2     | Deeltoets 2 | 0 |
|------------------------|-------------|---|
| Responsible Instructor | R. Oval     |   |
| Education Period       | 3           |   |
| Start Education        | 3           |   |
| Exam Period            | 3           |   |
| Course Language        | Dutch       |   |
| Collegerama            | Yes         |   |

| CTB2320-17 Toets 3     | Ethics CTB2320-17 | 0 |
|------------------------|-------------------|---|
| Responsible Instructor | R. Oval           |   |
| Education Period       | 3                 |   |
| Start Education        | 3                 |   |
| Exam Period            | 3                 |   |
| Course Language        | Dutch             |   |
| Collegerama            | Yes               |   |

| CTB2320-17 Toets 4     | Practicum | 2 |
|------------------------|-----------|---|
| Responsible Instructor | R. Oval   |   |
| Education Period       | 3         |   |
| Start Education        | 3         |   |
| Exam Period            | 3         |   |
| Course Language        | Dutch     |   |
| Collegerama            | Yes       |   |

| CTB2320-17 Toets 5     | Integrale Toets | 3 |
|------------------------|-----------------|---|
| Responsible Instructor | R. Oval         |   |
| Education Period       | 3               |   |
| Start Education        | 3               |   |
| Exam Period            | 3               |   |
| Course Language        | Dutch           |   |
| Collegerama            | Yes             |   |

| CTB2400                         | Numerical Mathematics                                                                                                                                                                                                                                                                                                                                                                                   | 3 |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| Responsible Instructor          | Prof.dr.ir. C. Vuik                                                                                                                                                                                                                                                                                                                                                                                     |   |
| Co-responsible for assignments  | Dr.ir. D. den Ouden-van der Horst                                                                                                                                                                                                                                                                                                                                                                       |   |
| Education Period                | 4                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| Start Education                 | 4                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| Exam Period                     | 4<br>5                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| Course Language                 | Dutch                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| Expected prior knowledge        | analysis, calculus, linear algebra, differential equations                                                                                                                                                                                                                                                                                                                                              |   |
| Parts                           | During the lectures the most important points are highlighted. Motivation for the various parts of the lectures/book are given. Exercises are discussed and explained. There is a lot of material available on the corresponding website for information, motivation, exercises and old exams.                                                                                                          |   |
| Summary                         | After the course the student is able to design a simple numerical method, analyses it and implement it. Furthermore, the student should be able to consider the results of numerical methods, analyse them and explain their behaviour. An oral and/or written report of these findings can be given by the student.                                                                                    |   |
| Education Method                | Lecture                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| Tags                            | Mathematics<br>Numeric Methods<br>Programming                                                                                                                                                                                                                                                                                                                                                           |   |
| Literature & Study Materials    | Numerical Methods for Ordinary Differential Equations<br>C. Vuik, F.J. Vermolen, M.B. van Gijzen and M.J. Vuik<br>TU Delft OPEN, Delft, 2023<br><a href="https://textbooks.open.tudelft.nl/textbooks/catalog/book/57">https://textbooks.open.tudelft.nl/textbooks/catalog/book/57</a>                                                                                                                   |   |
| Judgement                       | Grade of the written test is the final grade.<br><br>For more information on grading, see article 14 in the Rules and Guidelines (RGBE):<br><a href="https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations">https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations</a> |   |
| Permitted Materials during Exam | Only a standard calculator.                                                                                                                                                                                                                                                                                                                                                                             |   |
| Collegerama                     | Yes                                                                                                                                                                                                                                                                                                                                                                                                     |   |

| CTB2410                | Hydraulic Engineering | 5 |
|------------------------|-----------------------|---|
| Responsible Instructor | Ir. T.J. Zitman       |   |
| Education Period       | 4                     |   |
| Start Education        | 4                     |   |
| Exam Period            | 4<br>5                |   |
| Course Language        | Dutch                 |   |
| Collegerama            | No                    |   |

| CTB2420-17             | Hydrology                 | 5 |
|------------------------|---------------------------|---|
| Responsible Instructor | Prof.dr.ir. R. Uijlenhoet |   |
| Course Coordinator     | M. Zegel                  |   |
| Instructor             | Dr.ir. J.A. van der Werf  |   |
| Instructor             | Prof.dr.ir. R. Uijlenhoet |   |
| Instructor             | Dr. E. Mostert            |   |
| Instructor             | Dr.ir. A.M.J. Coenders    |   |
| Instructor             | Dr.ir. O.A.C. Hoes        |   |
| Instructor             | Dr.ir. R.J. van der Ent   |   |
| Instructor             | Dr.ir. J.M. Bloemendal    |   |
| Education Period       | 4                         |   |
| Start Education        | 4                         |   |
| Exam Period            | 4                         |   |
| Course Language        | Dutch                     |   |
| Collegerama            | Yes                       |   |

| <b>CTB2420-17 Toets 1</b> | <b>Deeltoets 1</b>        | <b>0</b> |
|---------------------------|---------------------------|----------|
| Responsible Instructor    | Prof.dr.ir. R. Uijlenhoet |          |
| Education Period          | 4                         |          |
| Start Education           | 4                         |          |
| Exam Period               | 4                         |          |
| Course Language           | Dutch                     |          |
| Collegerama               | Yes                       |          |

| <b>CTB2420-17 Toets 2</b> | <b>Deeltoets 2</b>        | <b>0</b> |
|---------------------------|---------------------------|----------|
| Responsible Instructor    | Prof.dr.ir. R. Uijlenhoet |          |
| Education Period          | 4                         |          |
| Start Education           | 4                         |          |
| Exam Period               | 4                         |          |
| Course Language           | Dutch                     |          |
| Collegerama               | Yes                       |          |

| <b>CTB2420-17 Toets 3</b> | <b>Ethics CTB2420-17</b>  | <b>0</b> |
|---------------------------|---------------------------|----------|
| Responsible Instructor    | Prof.dr.ir. R. Uijlenhoet |          |
| Education Period          | 4                         |          |
| Start Education           | 4                         |          |
| Exam Period               | 4                         |          |
| Course Language           | Dutch                     |          |
| Collegerama               | Yes                       |          |

| <b>CTB2420-17 Toets 4</b> | <b>Practicum</b>          | <b>0</b> |
|---------------------------|---------------------------|----------|
| Responsible Instructor    | Prof.dr.ir. R. Uijlenhoet |          |
| Education Period          | 4                         |          |
| Start Education           | 4                         |          |
| Exam Period               | 4                         |          |
| Course Language           | Dutch                     |          |
| Collegerama               | Yes                       |          |

| <b>CTB2420-17 Toets 5</b> | <b>Integrale Toets</b>    | <b>5</b> |
|---------------------------|---------------------------|----------|
| Responsible Instructor    | Prof.dr.ir. R. Uijlenhoet |          |
| Education Period          | 4                         |          |
| Start Education           | 4                         |          |
| Exam Period               | 4                         |          |
| Course Language           | Dutch                     |          |
| Collegerama               | Yes                       |          |

|                     |                                          |
|---------------------|------------------------------------------|
| <b>Year</b>         | <b>2024/2025</b>                         |
| <b>Organization</b> | <b>Civil Engineering and Geosciences</b> |
| <b>Education</b>    | <b>Bachelor Civil Engineering</b>        |

|                            |
|----------------------------|
| <b>CT BSc 3e jaar 2024</b> |
|----------------------------|

|                     |                                          |
|---------------------|------------------------------------------|
| <b>Year</b>         | <b>2024/2025</b>                         |
| <b>Organization</b> | <b>Civil Engineering and Geosciences</b> |
| <b>Education</b>    | <b>Bachelor Civil Engineering</b>        |

**CT BSc 3e jaar - minor**

|                     |                                          |
|---------------------|------------------------------------------|
| <b>Year</b>         | <b>2024/2025</b>                         |
| <b>Organization</b> | <b>Civil Engineering and Geosciences</b> |
| <b>Education</b>    | <b>Bachelor Civil Engineering</b>        |

|                               |
|-------------------------------|
| <b>CT BSc 3e jaar - major</b> |
|-------------------------------|

| CTB3000-16             | Bachelor Thesis                | 10 |
|------------------------|--------------------------------|----|
| Responsible Instructor | Ir. T.J. Zitman                |    |
| Responsible Instructor | Y. de las Heras                |    |
| Education Period       | 1<br>2<br>3<br>4               |    |
| Start Education        | 1<br>2<br>3<br>4               |    |
| Exam Period            | none                           |    |
| Course Language        | Dutch<br>English               |    |
| Course Contents        | Information available in Dutch |    |
| Collegerama            | No                             |    |

| CTB3000-16 Toets 1     | Ethics           | 0 |
|------------------------|------------------|---|
| Responsible Instructor | Ir. T.J. Zitman  |   |
| Education Period       | 1<br>2<br>3<br>4 |   |
| Start Education        | 1<br>2<br>3<br>4 |   |
| Exam Period            | none             |   |
| Course Language        | Dutch<br>English |   |
| Collegerama            | No               |   |

| CTB3000-16 Toets 2     | Presentation     | 10 |
|------------------------|------------------|----|
| Responsible Instructor | Ir. T.J. Zitman  |    |
| Education Period       | 1<br>2<br>3<br>4 |    |
| Start Education        | 1<br>2<br>3<br>4 |    |
| Exam Period            | none             |    |
| Course Language        | Dutch<br>English |    |
| Collegerama            | No               |    |



| CTB3310                                | Surveying and Mapping                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 4 |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr.ir. C.C.J.M. Tiberius                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Instructor</b>                      | Dr.ir. H. van der Marel                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Instructor</b>                      | R.H.C. Reudink                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Instructor</b>                      | Dr.ir. C.C.J.M. Tiberius                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Instructor</b>                      | Dr.ir. F.J. van Leijen                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/5/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Education Period</b>                | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Start Education</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Exam Period</b>                     | 3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Course Language</b>                 | Dutch                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Parts</b>                           | lectures (7x2 hours)<br>computer-based assignments/practical work/fieldwork (7x4 hours)<br>self-study                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Summary</b>                         | <p>The goal of this course is to convey the basic concepts and techniques of geodesy, remote sensing earth observation and data science, from a civil engineering application perspective. This covers both conventional as well as modern, also space-based, observation methods and techniques, and associated data processing, interpretation and visualization.</p> <p>The assignments and practical work are primarily about hands-on working with different sorts of equipment and data.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Course Contents</b>                 | <p>The lectures and practical exercises cover, each week, in parallel, generic basic material in surveying and mapping (techniques, methods, data processing and analysis, and background), as well as examples of applications in practice (design, realization and maintenance of constructions and infrastructure, traffic and transport, hydraulic and offshore engineering, geo-technical engineering, water-management and environmental engineering).</p> <ol style="list-style-type: none"> <li>1. Surveying for construction (tachymetry/total station, topographic surveying and stake-out, observation modeling, parameter estimation)</li> <li>2. Surveying for construction (design of measurement set-up, leveling, measurement uncertainty, variance, correlation, accuracy, error propagation, validation, interpolation, data quality control)</li> <li>3. Photogrammetry (central projection, relation photo geometry - terrain geometry, stereo-photo, mapping, 3D-reconstruction)</li> <li>4. Surveying for dynamic positioning (traffic, transport and navigation, infrastructure-construction), satellite-navigation systems: GPS/GNSS; 3D Digital Terrain Model (DTM) survey</li> <li>5. Coordinate and reference-systems for on and off-shore surveying (RD, ETRS89, ITRS (WGS84), NAP, geoid, MSL, LAT), offshore surveying (bathymetry by echo-sounding)</li> <li>6. Remote Sensing earth observation-techniques (satellite missions, satellite orbit, sensors/payloads, techniques, optical and radar remote sensing (radar deformation monitoring), laser-altimetry and laser-scanning, error propagation, limitations, new developments as atmospheric remote sensing), flooding assessment with 3D terrain modelling (AHN, TOP10NL and QGIS) and vegetation analysis (NDVI, based on Sentinel remote sensing data)</li> <li>7. Cartography (map-projection, visualization, overlays, 3D-objects, terrain representation and idealization, Google Earth/Maps), working with spatial data (visualization and analysis, layers, modeling, Geographic Information System (GIS), available geo-information, PDOK), spatial precipitation (rainfall) analysis, thematic mapping (QGIS)</li> </ol> |   |
| <b>Study Goals</b>                     | <p>Upon completion of this course, the student has:</p> <ul style="list-style-type: none"> <li>working knowledge about, and understanding of, reference systems, observation techniques (leveling, total station/tachymetry and laser-scanning), uncertainties in measurements, measurement processes, satellite systems for surveying and mapping (remote sensing earth observation, GPS/GNSS), Geographic Information Systems (GIS), cartography, and visualisation</li> <li>knowledge about, and understanding of the calculus of observation (waarnemingsrekening), and is able to apply this to actual problems (measurement data processing), and, analyse, evaluate, interpret and explain measures of quality of the survey, and the student is:</li> <li>able to advise on the choice of measurement techniques for civil engineering geo-problems, and to design a measurement set-up</li> <li>able to convert a measurement problem and set-up, into a mathematical model, to determine (estimate) the unknown parameters of interest based on the observed data, and to translate the outcome, including its quality (uncertainty), back into the application</li> <li>able to interpret, analyse and present spatial geo-information</li> </ul> <p>Next to applying course-related knowledge and developing accompanying skills, the practicals of this course contribute to development of an academic attitude, specifically:</p> <ul style="list-style-type: none"> <li>problem formulation and solving</li> <li>interpretation of data and data processing results (e.g. uncertainty and statistical significance)</li> <li>writing short reports</li> <li>reflect on outcomes of analyses (judgemental skills)</li> <li>reasoning and arguing</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Education Method</b>                | <p>weekly lectures (in Dutch)<br/>weekly (supervised) practical work and assignments<br/>self-study</p> <p>Lectures and practical work and assignments are on-campus. The computer-based assignments can be done on student's own PC/laptop</p> <p>This course is composed of:</p> <ul style="list-style-type: none"> <li>7x2 = 14 hours of lectures</li> <li>7x4 = 28 hours of practical work and assignments</li> <li>8x7 = 56 hours of self-study (7 hours per week)</li> <li>1x11 = 11 hours of preparation for exam</li> <li>1x3 = 3 hours of examination</li> <li>Total = 112 hours</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Course Relations</b>                | <ul style="list-style-type: none"> <li>- Inleiding Civiele Techniek CTB1120-17</li> <li>- Bouwplaats: CAD/BIM/BEM CTB1000</li> <li>- Bouwplaats CTB2000</li> <li>- Waterbouwkunde CTB2410</li> <li>- Hydrologie CTB2420-17</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Assessment</b>                      | <p>practical - toets1 (series of assignments)</p> <p>the assignments have to be all completed within one academic year - results of individual assignments can not be transferred to another academic year</p> <p>practical is assessed by pass/fail (a minimum of 6 out of 7 assignments need to be passed)</p> <p>written exam (closed book) - toets2</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |

|                                         |                                                                                                                                                                                                                                                                                                                                                    |
|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Expected prior Knowledge</b>         | mark for the written exam is the final grade for this course (once the practical has been completed with a pass)<br>Linear Algebra / Lineaire Algebra CTB1002<br>Probability and Statistics / Kansrekening en Statistiek CTB2200                                                                                                                   |
| <b>Academic Skills</b>                  | see with Study Goals                                                                                                                                                                                                                                                                                                                               |
| <b>Literature &amp; Study Materials</b> | Surveying and Mapping<br>Christian Tiberius, Hans van der Marel, René Reudink, Freek van Leijen<br>November 2022, TU Delft OPEN Textbook<br>ISBN 9789463664899 (e-book), 9789463664905 (softback), DOI 10.5074/T.2021.007<br>(chapters/sections indicated by [*] are not part of the material for the exam)                                        |
| <b>Judgement</b>                        | For more information on grading, see article 14 in the Rules and Guidelines (RGBE):<br><a href="https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations">https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations</a> |
| <b>Permitted Materials during Exam</b>  | pocket-calculator, and a drawing triangle (geo-driehoek)<br><br>The exam is closed-book.                                                                                                                                                                                                                                                           |
| <b>Collegerama</b>                      | No                                                                                                                                                                                                                                                                                                                                                 |

| <b>CTB3310 Toets 1</b>                  | <b>Practicum</b>                  | <b>0</b> |
|-----------------------------------------|-----------------------------------|----------|
| <b>Responsible Instructor</b>           | Dr.ir. C.C.J.M. Tiberius          |          |
| <b>Education Period</b>                 | 3                                 |          |
| <b>Start Education</b>                  | 3                                 |          |
| <b>Exam Period</b>                      | none                              |          |
| <b>Course Language</b>                  | English                           |          |
| <b>Course Contents</b>                  | see course-description of CTB3310 |          |
| <b>Study Goals</b>                      | see course-description of CTB3310 |          |
| <b>Education Method</b>                 | see course-description of CTB3310 |          |
| <b>Assessment</b>                       | see course-description of CTB3310 |          |
| <b>Expected prior Knowledge</b>         | see course-description of CTB3310 |          |
| <b>Academic Skills</b>                  | see course-description of CTB3310 |          |
| <b>Literature &amp; Study Materials</b> | see course-description of CTB3310 |          |
| <b>Judgement</b>                        | see course-description of CTB3310 |          |
| <b>Permitted Materials during Exam</b>  | see course-description of CTB3310 |          |
| <b>Collegerama</b>                      | No                                |          |

| <b>CTB3310 Toets 2</b>                  | <b>Tentamen</b>                   | <b>4</b> |
|-----------------------------------------|-----------------------------------|----------|
| <b>Responsible Instructor</b>           | Dr.ir. C.C.J.M. Tiberius          |          |
| <b>Education Period</b>                 | 3                                 |          |
| <b>Start Education</b>                  | 3                                 |          |
| <b>Exam Period</b>                      | 3<br>4                            |          |
| <b>Course Language</b>                  | English                           |          |
| <b>Course Contents</b>                  | see course-description of CTB3310 |          |
| <b>Study Goals</b>                      | see course-description of CTB3310 |          |
| <b>Education Method</b>                 | see course-description of CTB3310 |          |
| <b>Assessment</b>                       | see course-description of CTB3310 |          |
| <b>Expected prior Knowledge</b>         | see course-description of CTB3310 |          |
| <b>Academic Skills</b>                  | see course-description of CTB3310 |          |
| <b>Literature &amp; Study Materials</b> | see course-description of CTB3310 |          |
| <b>Judgement</b>                        | see course-description of CTB3310 |          |
| <b>Permitted Materials during Exam</b>  | see course-description of CTB3310 |          |
| <b>Collegerama</b>                      | No                                |          |

| CTB3320                       | Road and Railway Design        | 4 |
|-------------------------------|--------------------------------|---|
| <b>Responsible Instructor</b> | Prof.dr.ir. S.M.J.G. Erkens    |   |
| <b>Instructor</b>             | A. Jagadeesh                   |   |
| <b>Instructor</b>             | Prof.dr.ir. S.M.J.G. Erkens    |   |
| <b>Instructor</b>             | Dr. V.L. Markine               |   |
| <b>Instructor</b>             | Prof.dr.ir. R.P.B.J. Dollevoet |   |
| <b>Education Period</b>       | 3                              |   |
| <b>Start Education</b>        | 3                              |   |
| <b>Exam Period</b>            | 3                              |   |
|                               | 4                              |   |
| <b>Course Language</b>        | Dutch                          |   |
| <b>Collegerama</b>            | Yes                            |   |

|                     |                                          |
|---------------------|------------------------------------------|
| <b>Year</b>         | <b>2024/2025</b>                         |
| <b>Organization</b> | <b>Civil Engineering and Geosciences</b> |
| <b>Education</b>    | <b>Bachelor Civil Engineering</b>        |

**Overzicht Specialisatievakken. Student dient drie specialisatievakken te kiezen uit de lijsten hieronder**

|                     |                                          |
|---------------------|------------------------------------------|
| <b>Year</b>         | <b>2024/2025</b>                         |
| <b>Organization</b> | <b>Civil Engineering and Geosciences</b> |
| <b>Education</b>    | <b>Bachelor Civil Engineering</b>        |

**Sterk aanbevolen als voorbereiding op de MSc Environmental Engineering, kies 3 uit:**

| CTB3311                                 | Climate Impacts and Engineering                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 4 |
|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>           | Dr. A.A. Nuijens                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Instructor</b>                       | Dr. rer. nat. G. George                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Instructor</b>                       | Prof.dr. C.A. Katsman                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Contact Hours / Week</b><br>x/x/x/x  | 4 hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Education Period</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Start Education</b>                  | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Exam Period</b>                      | 3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Course Language</b>                  | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Course Contents</b>                  | Climate change is one of the dominant global long-term trends affecting the work of the civil engineer. It impacts the boundary conditions of coastal defence systems, off-shore engineering, water management and large infrastructure design. Optimal risk management and long-term sustainability are only possible when the potential consequences of climate change are taken into account. Not only mitigation and adaptation, but also the science behind climate change is an engineering problem, with several open-standing questions to solve. This course provides students with the current scientific knowledge of the climate system, of climate change, and the role of processes in the atmosphere, land and ocean in climate change. Furthermore, students will gain experience in the interpretation of climate variability and global and regional climate change relevant to engineering challenges.                                                                                           |   |
| <b>Study Goals</b>                      | <p>At the end of this class, students can:</p> <ol style="list-style-type: none"> <li>1. Explain the concept of climate sensitivity using basic principles of the Earth's radiation balance and analyze uncertainties in the feedbacks that play a role in climate sensitivity.</li> <li>2. Explain the roles of the ocean, atmosphere and land in climate and climate change.</li> <li>3. Explain how past and future climate change is modeled, monitored and analyzed.</li> <li>4. Interpret the different controls of the carbon cycle and assess the impact of engineering choices on atmospheric concentrations of greenhouse gases and associated climate change.</li> <li>5. Interpret variability in regional to local weather and weather extremes in light of global warming</li> <li>6. Analyze past and future variations in atmosphere, ocean or land processes in the presence of global warming and the significance of these (climate) changes on a specific civil engineering problem.</li> </ol> |   |
| <b>Education Method</b>                 | A combination of 7 weeks x 2 hours of oral lectures explaining the theory behind climate science and 7 weeks x 2 hours of practical "lab" sessions that include plenary assignments, guided group work and feedback sessions that help the students prepare for their group assignment.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Course Relations</b>                 | Related to (BSc) courses on hydrological/meteorological/hydraulic engineering. Overlap with the MSc course Climate Change and Ethics will be minimal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Assessment</b>                       | <p>Assessment is based on the ability of the student to apply theoretical information to 1) explain the role of the ocean and atmosphere in climate and climate change, 2) illustrate that knowledge in exercises, 3) interpret specific processes of the coupled atmosphere-ocean system and 4) evaluate the role of climate for a specific engineering challenge.</p> <p>Grade based on written exam (closed book) (60%) and written group report (40%). The minimum grade for the written exam is a 5. The resit in Q4 only applies to the written exam. The group report has to be completed in Q3.</p>                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Expected prior Knowledge</b>         | Basic level mathematics, mechanics, hydrology, fluid dynamics (Bachelor level).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Academic Skills</b>                  | Remembering and understanding basic theoretical concepts. Demonstrating this understanding through the application of theory. Analysis and interpretation of graphics. Critical thinking. Effective communication in discussions. Working in groups.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Literature &amp; Study Materials</b> | Lecture slides, a guided reader with book chapters and literature references, online climate analysis tools.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Judgement</b>                        | For more information on grading, see article 14 in the Rules and Guidelines (RGE):<br><a href="https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations">https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Permitted Materials during Exam</b>  | A calculator. This is a closed book exam.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Collegerama</b>                      | No                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |

| <b>CTB3311 Toets 1</b>                  | <b>Exam</b>                     | <b>2.5</b> |
|-----------------------------------------|---------------------------------|------------|
| <b>Responsible Instructor</b>           | Dr. A.A. Nuijens                |            |
| <b>Co-responsible for assignments</b>   | Prof.dr. C.A. Katsman           |            |
| <b>Co-responsible for assignments</b>   | Dr. rer. nat. G. George         |            |
| <b>Contact Hours / Week</b><br>x/x/x/x  | 0/0/0/4                         |            |
| <b>Education Period</b>                 | 3                               |            |
| <b>Start Education</b>                  | 3                               |            |
| <b>Exam Period</b>                      | 3<br>4                          |            |
| <b>Course Language</b>                  | English                         |            |
| <b>Course Contents</b>                  | See course description: CTB3311 |            |
| <b>Study Goals</b>                      | See course description: CTB3311 |            |
| <b>Education Method</b>                 | See course description: CTB3311 |            |
| <b>Assessment</b>                       | See course description: CTB3311 |            |
| <b>Expected prior Knowledge</b>         | See course description: CTB3311 |            |
| <b>Academic Skills</b>                  | See course description: CTB3311 |            |
| <b>Literature &amp; Study Materials</b> | See course description: CTB3311 |            |
| <b>Judgement</b>                        | See course description: CTB3311 |            |
| <b>Permitted Materials during Exam</b>  | See course description: CTB3311 |            |
| <b>Collegerama</b>                      | No                              |            |

| <b>CTB3311 Toets 2</b>                  | <b>Assignment</b>               | <b>1.5</b> |
|-----------------------------------------|---------------------------------|------------|
| <b>Responsible Instructor</b>           | Dr. A.A. Nuijens                |            |
| <b>Co-responsible for assignments</b>   | Prof.dr. C.A. Katsman           |            |
| <b>Co-responsible for assignments</b>   | Dr. rer. nat. G. George         |            |
| <b>Contact Hours / Week</b><br>x/x/x/x  | 0/0/0/4                         |            |
| <b>Education Period</b>                 | 3                               |            |
| <b>Start Education</b>                  | 3                               |            |
| <b>Exam Period</b>                      | 3                               |            |
| <b>Course Language</b>                  | English                         |            |
| <b>Course Contents</b>                  | See course description: CTB3311 |            |
| <b>Study Goals</b>                      | See course description: CTB3311 |            |
| <b>Education Method</b>                 | See course description: CTB3311 |            |
| <b>Assessment</b>                       | See course description: CTB3311 |            |
| <b>Expected prior Knowledge</b>         | See course description: CTB3311 |            |
| <b>Academic Skills</b>                  | See course description: CTB3311 |            |
| <b>Literature &amp; Study Materials</b> | See course description: CTB3311 |            |
| <b>Judgement</b>                        | See course description: CTB3311 |            |
| <b>Permitted Materials during Exam</b>  | See course description: CTB3311 |            |
| <b>Collegerama</b>                      | No                              |            |

| CTB3350                                 | Open Channel Flow                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 4 |
|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>           | Dr.ir. R.J. Labeur                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Instructor</b>                       | J.A. Arriaga Garcia                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Contact Hours / Week</b><br>x/x/x/x  | 0/0/6/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Education Period</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Start Education</b>                  | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Exam Period</b>                      | 3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Course Language</b>                  | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Expected prior knowledge</b>         | basic fluid mechanics, ordinary and partial differential equations, complex numbers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Parts</b>                            | lectures<br>MAPLE TA tests                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Course Contents</b>                  | Shallow-water equations, preliminary analyses and classification of waves;<br>Elementary wave equation, solution method and analyses; Translatory waves, interaction with control structures, tidal bore; Method of characteristics, sub- and super-critical flow, transients in closed conduits; Tidal basins, influence resistance and inertia, resonance; Harmonic method, Lorentz-linearisation, tidal propagation in large systems; High water waves, kinematic waves and diffusion model, hysteresis.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Study Goals</b>                      | After mastering the course the student should be able to identify, analyze and compute various wave phenomena encountered in hydraulic engineering and water management that are governed by the one-dimensional shallow water equations. This relies on selective schematization and choosing adequate solution strategies which involves: <ul style="list-style-type: none"> <li>- understanding the role of the hydrostatic pressure approximation</li> <li>- classification and analyses of waves based on resistance versus inertia</li> <li>- understanding the basic structure of elementary wave solutions</li> <li>- addressing initial and boundary conditions, (partial) reflection</li> <li>- graphical construction of solutions using the method of characteristics</li> <li>- analyses of small basins with discrete storage, resistance and inertia</li> <li>- analyses of traveling and standing waves in large basins including resistance</li> <li>- analyses of high water waves with dominant resistance, kinematic waves, diffusion model</li> </ul> |   |
| <b>Education Method</b>                 | lectures<br>MAPLE TA tests (homework)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Course Relations</b>                 | fluid mechanics, dynamics, calculus, water management, hydraulic structures, computational modeling of flow and transport                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Literature and Study Materials</b>   | Textbook "Unsteady Flow in Open Channels", Jurjen Battjes and Robert Jan Labeur, Cambridge University Press, ed. 2017; available through, amazon.com, bol.com, TUD library (digital copy) <p>Available on BrightSpace:</p> <ul style="list-style-type: none"> <li>- lecture slides</li> <li>- answers to problem sets</li> <li>- compilation of assignments</li> <li>- various Python scripts and videos</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Assessment</b>                       | written exam (summative)<br>online ANS tests for self assessment (not compulsory)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Permitted Materials during Tests</b> | pen, pencils, ruler, eraser<br>(graphical) non-programmable calculator as described in the examination regulations.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Elective</b>                         | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Tags</b>                             | Fluid Mechanics<br>Physics<br>Water Engineering<br>Water management                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Expected prior Knowledge</b>         | introduction environmental fluid mechanics<br>dynamics<br>ordinary- and partial differential equations<br>complex numbers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Academic Skills</b>                  | problem analysis<br>reduction of complex systems<br>analyses of system behaviour                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Literature &amp; Study Materials</b> | Textbook "Unsteady Flow in Open Channels", Jurjen Battjes and Robert Jan Labeur, Cambridge University Press, ed. 2017; available through VSSD, amazon.com, bol.com <p>Available on BrightSpace:</p> <ul style="list-style-type: none"> <li>- lecture slides</li> <li>- answers to problem sets</li> <li>- compilation of assignments</li> <li>- various MatLab scripts and videos</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Judgement</b>                        | Final grade is grade of written exam<br>online ANS tests are for self assessment only <p>For more information on grading, see article 14 in the Rules and Guidelines (RGBE):<br/><a href="https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations">https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Permitted Materials during Exam</b>  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Collegerama</b>                      | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |



| CTB3360                                 | Water System Analysis                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 4 |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>           | Dr.ir. G.H.W. Schoups                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Contact Hours / Week</b><br>x/x/x/x  | 4/0/4/0<br><br>Hoorcollege: 2<br>Werkcollege: 2<br>Practica: 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Education Period</b>                 | 1<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Start Education</b>                  | 1<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Exam Period</b>                      | 1<br>2<br>3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Course Language</b>                  | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Expected prior knowledge</b>         | Fluid mechanics<br>Introduction water management<br>Introduction hydrology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Course Contents</b>                  | <p>Central topic of the course is analysis and design of natural and man-made water systems (canals, rivers, agricultural fields, polders, irrigation systems, soils, aquifers) based on hydraulic and hydrological knowledge. Common flow processes encountered in practice are discussed, including open-channel flow, hydraulic structures, soil water flow, and groundwater flow. In particular, the course focuses on:</p> <p>(1) unified treatment of the physics of water flow in surface and subsurface systems based on the principles of mass and momentum conservation</p> <p>(2) basic numerical methods for solving the relevant flow equations</p> <p>(3) application of theory to existing systems from around the world.</p> <p>As such, the course consolidates and builds on prior knowledge obtained in other courses, and prepares for more advanced topics in the MSc Watermanagement program.</p>                                               |   |
| <b>Study Goals</b>                      | <p>The overall goal is to analyze and quantify water flow through natural and man-made systems such as canals, rivers, agricultural fields, polders, irrigation systems, soils, and aquifers. This requires the following:</p> <p>quantitative understanding of flow processes commonly encountered in practice, including open-channel flow, hydraulic structures, soil water flow, and groundwater flow</p> <p>translate a given practical flow situation into a mathematical model of water flow in the system</p> <p>apply basic numerical techniques for solving the resulting mathematical models</p> <p>interpret computed results and draw conclusions about the functioning of the water system under consideration, and, if applicable, formulate control actions to improve the situation</p> <p>Study load:<br/>Lectures: 7 x 2 hours<br/>Assignments: 6 x 10 hours<br/>Self study: 7 x 2 hours<br/>Exam preparation: 8 hours<br/>Final exam: 3 hours</p> |   |
| <b>Education Method</b>                 | Lectures and Assignments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Course Relations</b>                 | Water management, Hydrology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Literature and Study Materials</b>   | All study materials (lecture slides) will be provided on BrightSpace                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Assessment</b>                       | Weekly computer assignments and final exam, with a split of 40/60 or 60/40, whichever yields the higher final grade. You need a grade of 5.75 or higher on the exam.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Permitted Materials during Tests</b> | Calculator and A4 paper with formulas prepared by the student                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Expected prior Knowledge</b>         | Fluid mechanics<br>Introduction water management<br>Introduction hydrology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Academic Skills</b>                  | Analytical thinking<br>Report writing<br>Problem solving                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Literature &amp; Study Materials</b> | All study materials (lecture slides) will be provided on BrightSpace                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Judgement</b>                        | For more information on grading, see article 14 in the Rules and Guidelines (RGBE):<br><a href="https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations">https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Permitted Materials during Exam</b>  | Calculator and A4 paper with formulas prepared by the student                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Collegerama</b>                      | No                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |

| CTB3365-16                              | Introduction to Water Treatment                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 4 |
|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>           | Prof.dr.ir. J.B. van Lier                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Instructor</b>                       | Prof.dr.ir. L.C. Rietveld                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Instructor</b>                       | Prof.dr.ir. M.K. de Kreuk                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Contact Hours / Week</b><br>x/x/x/x  | 0/0/6/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Education Period</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Start Education</b>                  | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Exam Period</b>                      | 3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Course Language</b>                  | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Parts</b>                            | Drinking Water Treatment<br>Wastewater Treatment                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Summary</b>                          | Understand the process units and make a process design of a drinking water and wastewater treatment plant                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Course Contents</b>                  | This course deals with the basic principles and design aspects of sanitary engineering infrastructure. This comprises: drinking water quality and treatment and wastewater characteristics and treatment                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Study Goals</b>                      | <p>At the end of this course you will be able to:</p> <ol style="list-style-type: none"> <li>1. Identify the drinking water quality parameters to be improved and explain what treatment chain or scheme is needed</li> <li>2. Calculate the dimensions of the drinking water treatment processes</li> <li>3. Draw drinking water treatment schemes</li> <li>4. Identify the wastewater quality parameters to be improved and explain what treatment train is needed</li> <li>5. Calculate the dimensions of the wastewater treatment processes</li> <li>6. Draw wastewater treatment schemes</li> </ol> <p>The course consists of 4 ECTS which means a study load of 112 hours. This is divided into:</p> <ul style="list-style-type: none"> <li>42 hours classes (6 hours per week during 7 weeks)</li> <li>42 hours preparation and self study</li> <li>24 hours preparation on exams</li> <li>4 hours exams</li> </ul> |   |
| <b>Education Method</b>                 | <p>The theory is available in lecture notes/books, slides and collegerama on BrightSpace. The course is also available as Massive Online Open Course (MOOC) on the EdX platform.</p> <p>Oral lectures will highlight the most important aspects, together with exercises and exam preparation. One block lecture will consist of excursions. The course will be given on-campus or online (depending on situation)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Course Relations</b>                 | BSc course Water Management (in Dutch)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Assessment</b>                       | <p>Online and On-campus assessment forms.</p> <p>The testing is done by two written partial exams. The first partial exam is done in week 5 and the second in week 8-10.</p> <p>The partial exams consist of 3 parts testing:</p> <ul style="list-style-type: none"> <li>- Knowledge</li> <li>- Calculations of water treatment processes</li> <li>- Drawing of treatment schemes</li> </ul> <p>50% exam drinking water treatment<br/>50% exam wastewater treatment</p> <p>The minimum mark for both partial exams is 5.0 for passing.</p> <p>Re-exam includes both partial exams. Partial marks will not be kept.</p>                                                                                                                                                                                                                                                                                                     |   |
| <b>Expected prior Knowledge</b>         | Introduction to Watermanagement, CTB2011                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Academic Skills</b>                  | Critical analytical thinking, interpretation, problem solving                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Literature &amp; Study Materials</b> | <p>Lecture notes. Down loadable from BrightSpace</p> <p>Book. Drinking water-principles and practices P.J.de Moel, J.Q.J.C. Verberk, J.C.van Dijk, World Science. Available at book stores and secretary sanitary engineering, room 4.55</p> <p>Optional: Wastewater Engineering: MetCalf &amp; Eddy: Treatment and Reuse. 4th ed. McGraw-Hill publ.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Judgement</b>                        | <p>For more information on grading, see article 14 in the Rules and Guidelines (RGBE):<br/> <a href="https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations">https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations</a> </p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Permitted Materials during Exam</b>  | <p>Calculator as described in the exam regulations</p> <p>Drawing material</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Collegerama</b>                      | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |

| CTB3390                                 | Mechanics and Transport by Flow through Poreus Media                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 4 |
|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>           | S. Muraro                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Course Coordinator</b>               | S. Muraro                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Contact Hours / Week</b><br>x/x/x/x  | 0/0/4/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Education Period</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Start Education</b>                  | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Exam Period</b>                      | 3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Course Language</b>                  | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Course Contents</b>                  | <p>This course is jointly given with AESB3340.</p> <p>In this course, students acquire a fundamental basis for one- and two-phase flow in porous media. This course is an essential basis for all tracks within the MSc Applied Earth Sciences as well as the MSc track in Geotechnical Engineering in the MSc Civil Engineering. This course is highly recommended for students interested in Hydraulic Engineering as well as Hydro-geology and Water Management.</p> <p>The course will address the following topics:</p> <ul style="list-style-type: none"> <li>- basic principles for groundwater flow under saturated conditions (1D, 2D and 3D, both stationary as well as non-stationary flow);</li> <li>- (potential) energy and its relationship with pressure and head;</li> <li>- capillary pressure and unsaturated flow (2-phase flow in a porous medium filled with water and air);</li> <li>- concepts of water and air content, saturation, hydraulic conductivity, relative permeability, and residual saturation;</li> <li>- water retention curves (van Genuchten, Brooks-Corey, and Gardner equations);</li> <li>- the impact of flow on soil mechanics, highlighting the connection with the parallel course of soil mechanics.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Study Goals</b>                      | <p>At the end of this course you will be able to:</p> <ul style="list-style-type: none"> <li>- explain the origins of groundwater flow;</li> <li>- derive the governing equations of the groundwater flow with appropriate initial and boundary conditions for saturated soils and unsaturated soils;</li> <li>- solve laboratory and field steady-state and transient problems for saturated (1D, 2D) and unsaturated (1D) flow by programming a Python/Matlab script;</li> <li>- discuss the quality and the limitations of the model results.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Education Method</b>                 | <p>The course is structured along with weekly lectures and online assignments. The aim of this structure is to provide students with feedback on their progress. The content is structured with a study guide which is based on topics discussed in a number of standard handbooks.</p> <p>- Self-study<br/>The key to passing this course is self-study. The heart of the course is a study guide. In this guide, all topics are specified based on assignments that the students are required to make. The study guide points to the relevant chapters and paragraphs in a number of handbooks that the students need to study in order to be able to solve the assignments. There are several very good reference books available that cover the content of this course.</p> <p>- Lectures<br/>The students are expected to prepare for each lecture by making assignments and reading the required background material. The course is organised along with online assignments, activities integrated into Brightspace. These activities are a series of test assignments with which the students can evaluate their progress. Students are expected to upload questions to Brightspace about the assigned work or any subjects they have difficulties with. During the lectures, these questions will be addressed.</p> <p>- Asynchronous feedback and dialog<br/>Using the Discussion Board in the Brightspace environment, students are expected to collaborate in order to identify questions and to help each other complete the assignments. These tools are also used to give and get feedback on intermediate results (i.e. the assignments) but also on lectures etc.</p> |   |
| <b>Course Relations</b>                 | <p>This course is closely related with Soil Mechanics (CTB2310) and Physical Transport Phenomena (AESB2320). The course is fundamental to the Geo-engineering field and integrates different aspects with respect to the flow of water through porous media.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Literature and Study Materials</b>   | <p>Study guide, lecture notes, material on Brightspace, links to several good handbooks.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Assessment</b>                       | <p>Final exam in week 10. The exam consists of a series of questions that are similar to the assignments prepared for and discussed during the lectures.</p> <p>The online assignments given during the course are compulsory activities and enter in the final summative assessment (after the final exam is passed).</p> <p>The final grade is calculated as:<br/>70% exam + 30% Online Activities</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Permitted Materials during Tests</b> | <p>No intermediate tests will be given.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Contact</b>                          | <p>Dr. Stefano Muraro<br/>Email S.Muraro@tudelft.nl<br/>Phone +31 15 27 83327</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Expected prior Knowledge</b>         | <p>All Math, Mechanics 1, Soil Mechanics</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Academic Skills</b>                  | <p>Knowledge, students:</p> <ul style="list-style-type: none"> <li>- know the concepts and assumptions which form the basis of Darcy's Law;</li> <li>- are able to apply Darcy's law in a range of different applications;</li> <li>- understand how to simplify the 3D flow equation to 2D or 1D;</li> <li>- are able to explain when the assumptions used in the reduction of dimensionality are applicable;</li> <li>- understand the concepts of capillary pressure and unsaturated flow;</li> <li>- understand and be able to use the van Genuchten, Brooks-Corey, and Gardner equations;</li> </ul> <p>Skills, students:</p> <ul style="list-style-type: none"> <li>- are able to derive quantitative solutions to a range of engineering problems using analytical solutions;</li> <li>- have developed skills to share information and knowledge in an online environment;</li> <li>- have developed skills to help others in an online environment;</li> <li>- have developed the skill to use Python/Matlab to assess simple numerical and analytical solutions to flow and transport problems;</li> <li>- will have the skill to integrate transport of solutes with the groundwater flow problems in order to analyse groundwater contamination problems;</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                      |   |

|                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Literature &amp; Study Materials</b> | <p>Attitude, students:</p> <ul style="list-style-type: none"> <li>- will have developed an appreciation of the intrinsic value of the sub-surface for society, ecosystem etc.</li> <li>- will have developed an open attitude for collaboration in an online environment.</li> </ul> <p>All course materials are given on the Brightspace site.<br/>The course is based on the following books:<br/>Groundwater Science, 2013, Charles R. Fitts, second edition, Academic Press, Elsevier.<br/>Unsaturated Soil Mechanics, 2004, Lu, Ning; Likos, William J., John Wiley &amp; Sons.</p> |
| <b>Judgement</b>                        | <p>You are strongly advised to buy the book by Fitts. For Lu and Likos, we only use a small number of chapters. The relevant material will be provided during the class.</p> <p>For more information on grading, see article 14 in the Rules and Guidelines (RGBE):<br/><a href="https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations">https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations</a></p>                                                  |
| <b>Permitted Materials during Exam</b>  | The exam is an open-book exam, the student is allowed to bring all books, lecture notes, and worked examples. The student is allowed to use an electronic calculator. The exam is an individual assessment and no collaboration is allowed.                                                                                                                                                                                                                                                                                                                                              |
| <b>Collegerama</b>                      | No                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

| <b>CTB3415</b>                | <b>Water Management Research</b> | <b>4</b> |
|-------------------------------|----------------------------------|----------|
| <b>Responsible Instructor</b> | Dr.ir. M.W. Ertsen               |          |
| <b>Education Period</b>       | 4                                |          |
| <b>Start Education</b>        | 4                                |          |
| <b>Exam Period</b>            | 4                                |          |
| <b>Course Language</b>        | English                          |          |
| <b>Collegerama</b>            | No                               |          |

|                     |                                          |
|---------------------|------------------------------------------|
| <b>Year</b>         | <b>2024/2025</b>                         |
| <b>Organization</b> | <b>Civil Engineering and Geosciences</b> |
| <b>Education</b>    | <b>Bachelor Civil Engineering</b>        |

**Sterk aanbevolen als voorbereiding op de MSc Civil Engineering, kies 3 uit:**

| CTB3330                                | Structural Mechanics 4 | 4 |
|----------------------------------------|------------------------|---|
| <b>Responsible Instructor</b>          | Ir. J.W. Welleman      |   |
| <b>Instructor</b>                      | T.R. van Woudenberg    |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/6/0                |   |
| <b>Education Period</b>                | 3                      |   |
| <b>Start Education</b>                 | 3                      |   |
| <b>Exam Period</b>                     | 3<br>4                 |   |
| <b>Course Language</b>                 | English                |   |
| <b>Collegerama</b>                     | No                     |   |

| CTB3335                                 | Concrete Structures 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 4 |
|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>           | Dr. M. Lukovi                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Contact Hours / Week</b><br>x/x/x/x  | 0/0/4/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Education Period</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Start Education</b>                  | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Exam Period</b>                      | 3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Course Language</b>                  | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Expected prior knowledge</b>         | Concrete and Steel Structures<br>Structural Mechanics I<br>Structural Mechanics II<br>Structural Mechanics III                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Parts</b>                            | The course Concrete Structures 2 (CTB3335) contains two theoretical parts:<br>Reinforced concrete. This part consists of the theories related to slab design (one way, two way spanning slabs and punching shear), crack width control, torsion design.<br>Statically determinate prestressed girders. Design of prestressed concrete girders, prestress losses, capacity and detailing.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Summary</b>                          | This course is to give an introduction to the basic knowledge on how to design and assess the serviceability and safety of reinforced and statically determinate prestressed structures.<br>This is a strongly recommended course for students that plan to choose one of following tracks in the Civil Engineering MSc Program at TU Delft: Structural Engineering, Construction Materials, or Hydraulic and Offshore Structures.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Course Contents</b>                  | Design of reinforced concrete members under torsion.<br>Crack width calculation based on the tensile member model in both crack formation stage & stabilized cracking stage.<br>Design of reinforced concrete slabs spanning in one and two directions. Including different calculation methods for the internal forces, such as elastic analysis and equilibrium method for slabs with beams and flat slabs.<br>Punching shear resistance. Theory and application.<br>Introduction about the differences in the design of reinforced concrete and prestressed concrete.<br>Design of statically determinate prestressed concrete girders. Principles and materials, and prestressing as an external load.<br>Prestress losses: friction, slip, creep, shrinkage, relaxation.<br>Bending moment capacity, ultimate limit state.<br>Detailing, introduction of prestressing forces.                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Study Goals</b>                      | After completion of this course, students should be able to:<br><br>Calculate the torsional capacity of a reinforced concrete member, and identify the design limit state when a torsion check necessary.<br>Identify the different stage of cracking for a reinforced concrete structure, and calculate the crack width of a member under tension or bending.<br>Determine cross-sections and rebar configurations of one way spanning slabs, two way spanning slabs and flat slabs, based on theory of elasticity or equilibrium method.<br>Calculate the punching shear capacity of a flat slab, and decide the appropriate strengthening approach when the punching shear capacity is not sufficient.<br>Determine cross-sections and tendon configurations of pre-stressed concrete girders and explain the influence of the design on internal forces and stresses.<br>Calculate prestress losses.<br>Calculate the capacity of prestressed concrete girders in ULS.<br><br>This course is 4 ECTS, which corresponds to a study load of 112 hours. The presented study load is indicative and might vary based on the prior knowledge of the student.<br>Lectures 28 hours<br>Self-study 60 hours<br>Preparation for the exam 21 hours<br>Exam 3 hours |   |
| <b>Education Method</b>                 | The education methods exist of lectures in combination with self-study. Furthermore, students can actively participate with elective, formative questions during the lecture series.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Course Relations</b>                 | Concrete Structures 2 is used by: (1) Structural Engineering Track base module, (2) Modules B3 (Concrete Structures) and B4 (Building Engineering) within the Structural Engineering Track, and (3) Modules in Hydraulic and Offshore Structures.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Literature and Study Materials</b>   | Reinforced concrete: Chapters 11, 14 and 15, namely Torsion, Slabs/punching shear and Crack width control (material is available on the Brightspace)<br>Textbook Prestressed concrete (most recent version: 2019, available on the Brightspace) with following chapters:<br>Chapters 1 to 4 inclusive.<br>Chapter 6 sections 6.1, 6.2.1, 6.2.2, 6.5 and the example from 6.6.2.<br>Chapter 7 sections 7.1 and 7.2.<br>Chapter 10 sections 10 to 10.2.3 inclusive.<br>Booklet examples from exams:<br>Examples reinforced concrete.<br>Examples prestressed concrete.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Assessment</b>                       | The learning objectives will be assessed based on the exam (consisting of questions to perform calculations, or derive certain equation). The course is finalized with a 3 hour written examination.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Permitted Materials during Tests</b> | Calculator as described by the examination regulations<br>1 - A4 with hand written notes (one side only).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Expected prior Knowledge</b>         | Know how to draw a moment/shear diagram of beams with different boundary conditions;<br>Understand the difference between Ultimate Limit State and Serviceability Limit State.<br>Be able to design the cross section and reinforcement of a reinforced concrete beam for bending and shear.<br>Have the basic knowledge of the material properties of concrete and steel.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Academic Skills</b>                  | Analysis, Application, Literature survey                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Literature &amp; Study Materials</b> | See above                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Judgement</b>                        | See above                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
|                                         | For more information on grading, see article 14 in the Rules and Guidelines (RGBE):<br><a href="https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations">https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Permitted Materials during Exam</b>  | See above                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Collegerama</b>                      | No                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |

| CTB3340-15                              | Building Structures 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 4 |
|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>           | Ir. S. Pasterkamp                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Instructor</b>                       | J. Cupa                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Instructor</b>                       | A.C.B. Schuurman                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Instructor</b>                       | Dr.ir. G.J.P. Ravenshorst                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Contact Hours / Week</b><br>x/x/x/x  | 0/0/4/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Education Period</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Start Education</b>                  | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Exam Period</b>                      | 3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Course Language</b>                  | Dutch                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Expected prior knowledge</b>         | Ontwerpen van constructies en funderingen<br>Beton en staalconstructies                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Parts</b>                            | -Workshops special aspects low rise structures<br>-self study of reader and written test<br>-design exercise and presentation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Course Contents</b>                  | The course consists of 10 modules:<br>Introduction low rise<br>Low rise steel<br>Low rise in timber<br>Low rise in prefab concrete<br>Low rise in structural glass<br>Floors and foundations<br>Introduction housing<br>Housing in situ<br>Housing in masonry<br>Housing in timber                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Study Goals</b>                      | Demonstrate understanding of various aspects of low rise buildings and be able to design low rise building structures, with a specific focus on:<br>Functional, structural and construction requirements for housing and industrial buildings<br>Relationship between load bearing structure and finishing structure for roof, facade and ground floor for housing and industrial buildings<br>Steel and concrete elements on intermediate level (know requirements related to transport, assemblage and maintenance, design and calculate of structures with some complexity, select suitable detail and construction options)<br>Timber elements on introductory level (design and calculate simple structures, select suitable detail and construction options, understand thermal behaviour)<br>Glass or masonry elements on introductory level (understand relevant material properties, apply relevant safety procedure, design and calculate simple structures, apply suitable detail options)<br>Stability systems (braced and unbraced structures), with attention to second order forces (understand differences, calculate frameworks with second order effect)<br>Foundations on intermediate level (understand various ways of foundations, be able to calculate stiffness of various foundations)<br>Specific aspects of ground floors for industrial buildings and housing (know various types of ground floors with advantages and disadvantages, apply suitable measures to deal with cracking of concrete ground floors) |   |
| <b>Education Method</b>                 | Workshops and design exercise                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Course Relations</b>                 | Building Structures 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Literature and Study Materials</b>   | Reader Building structures 1, additional study material from workshops to be appointed.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Assessment</b>                       | The final grade is made up of a grade for the exam (50%) and a grade for the group assignment (50%). The minimum grade for each part is 5.0.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Elective</b>                         | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Tags</b>                             | Building Engineering                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Expected prior Knowledge</b>         | Ontwerpen van constructies en funderingen.<br>Beton en staalconstructies                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Academic Skills</b>                  | Integral design approach for various materials                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Literature &amp; Study Materials</b> | Reader building structures 1, other study material provided on Brightspace                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Judgement</b>                        | For more information on grading, see article 14 in the Rules and Guidelines (RGE):<br><a href="https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations">https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Permitted Materials during Exam</b>  | Calculator+quick reference                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Collegerama</b>                      | No                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |



| CTB3350                                 | Open Channel Flow                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 4 |
|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>           | Dr.ir. R.J. Labeur                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Instructor</b>                       | J.A. Arriaga Garcia                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Contact Hours / Week</b><br>x/x/x/x  | 0/0/6/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Education Period</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Start Education</b>                  | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Exam Period</b>                      | 3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Course Language</b>                  | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Expected prior knowledge</b>         | basic fluid mechanics, ordinary and partial differential equations, complex numbers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Parts</b>                            | lectures<br>MAPLE TA tests                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Course Contents</b>                  | Shallow-water equations, preliminary analyses and classification of waves;<br>Elementary wave equation, solution method and analyses; Translatory waves, interaction with control structures, tidal bore; Method of characteristics, sub- and super-critical flow, transients in closed conduits; Tidal basins, influence resistance and inertia, resonance; Harmonic method, Lorentz-linearisation, tidal propagation in large systems; High water waves, kinematic waves and diffusion model, hysteresis.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Study Goals</b>                      | After mastering the course the student should be able to identify, analyze and compute various wave phenomena encountered in hydraulic engineering and water management that are governed by the one-dimensional shallow water equations. This relies on selective schematization and choosing adequate solution strategies which involves: <ul style="list-style-type: none"> <li>- understanding the role of the hydrostatic pressure approximation</li> <li>- classification and analyses of waves based on resistance versus inertia</li> <li>- understanding the basic structure of elementary wave solutions</li> <li>- addressing initial and boundary conditions, (partial) reflection</li> <li>- graphical construction of solutions using the method of characteristics</li> <li>- analyses of small basins with discrete storage, resistance and inertia</li> <li>- analyses of traveling and standing waves in large basins including resistance</li> <li>- analyses of high water waves with dominant resistance, kinematic waves, diffusion model</li> </ul> |   |
| <b>Education Method</b>                 | lectures<br>MAPLE TA tests (homework)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Course Relations</b>                 | fluid mechanics, dynamics, calculus, water management, hydraulic structures, computational modeling of flow and transport                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Literature and Study Materials</b>   | Textbook "Unsteady Flow in Open Channels", Jurjen Battjes and Robert Jan Labeur, Cambridge University Press, ed. 2017; available through, amazon.com, bol.com, TUD library (digital copy) <p>Available on BrightSpace:</p> <ul style="list-style-type: none"> <li>- lecture slides</li> <li>- answers to problem sets</li> <li>- compilation of assignments</li> <li>- various Python scripts and videos</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Assessment</b>                       | written exam (summative)<br>online ANS tests for self assessment (not compulsory)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Permitted Materials during Tests</b> | pen, pencils, ruler, eraser<br>(graphical) non-programmable calculator as described in the examination regulations.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Elective</b>                         | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Tags</b>                             | Fluid Mechanics<br>Physics<br>Water Engineering<br>Water management                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Expected prior Knowledge</b>         | introduction environmental fluid mechanics<br>dynamics<br>ordinary- and partial differential equations<br>complex numbers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Academic Skills</b>                  | problem analysis<br>reduction of complex systems<br>analyses of system behaviour                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Literature &amp; Study Materials</b> | Textbook "Unsteady Flow in Open Channels", Jurjen Battjes and Robert Jan Labeur, Cambridge University Press, ed. 2017; available through VSSD, amazon.com, bol.com <p>Available on BrightSpace:</p> <ul style="list-style-type: none"> <li>- lecture slides</li> <li>- answers to problem sets</li> <li>- compilation of assignments</li> <li>- various MatLab scripts and videos</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Judgement</b>                        | Final grade is grade of written exam<br>online ANS tests are for self assessment only <p>For more information on grading, see article 14 in the Rules and Guidelines (RGBE):<br/><a href="https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations">https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Permitted Materials during Exam</b>  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Colleggerama</b>                     | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |

| CTB3355                                 | Hydraulic Structures                                                                                                      | 4 |
|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>           | Dr.ing. M.Z. Voorendt                                                                                                     |   |
| <b>Contact Hours / Week</b><br>x/x/x/x  | 0/0/6/0                                                                                                                   |   |
| <b>Education Period</b>                 | 3                                                                                                                         |   |
| <b>Start Education</b>                  | 3                                                                                                                         |   |
| <b>Exam Period</b>                      | 3<br>4                                                                                                                    |   |
| <b>Course Language</b>                  | Dutch                                                                                                                     |   |
| <b>Course Relations</b>                 |                                                                                                                           |   |
| <b>Literature and Study Materials</b>   |                                                                                                                           |   |
| <b>Permitted Materials during Tests</b> |                                                                                                                           |   |
| <b>Elective</b>                         | Yes                                                                                                                       |   |
| <b>Tags</b>                             | Broad<br>Design<br>Drawing<br>Fluid Mechanics<br>Intensive<br>Soil Mechanics<br>Structural Mechanics<br>Water Engineering |   |
| <b>Judgement</b>                        |                                                                                                                           |   |
| <b>Collegerama</b>                      | No                                                                                                                        |   |

| CTB3370-18                              | Geometrical Design of Roads and Railways                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 4 |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>           | Dr.ir. H. Farah                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Instructor</b>                       | Dr.ir. A.M. Salomons                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Instructor</b>                       | Prof.dr. Z. Li                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Contact Hours / Week</b><br>x/x/x/x  | 0.0.10.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Education Period</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Start Education</b>                  | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Exam Period</b>                      | 3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Course Language</b>                  | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Course Contents</b>                  | Geometric design of roads and railways: Cross sections, horizontal and vertical alignment, integration of the horizontal and vertical alignments;<br>Intersection design, roundabouts and turbo-roundabouts, interchanges, discontinuities;<br>Design for pedestrians and cyclists;<br>Railroad categories, and characteristic of railroad traffic;<br>Functional design of railway-infrastructure;<br>Design of trams in urban areas;<br>Human factors and road systems design;<br>Using AutoCAD and Civil3D;                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Study Goals</b>                      | At the end of this course you will be able to:<br>Apply design guidelines and design methodologies for designing the geometric alignment of roads and railways.<br>Explain the relationship between road design and road user behaviour.<br>Interpret and analyse characteristics of railway infrastructure elements.<br>Design railways and tram lines.<br>Display roads and railways design using AutoCAD and Civil3D.<br>Evaluate the implications of different design alternatives on road user behaviour, traffic safety and traffic operation.<br>Think critically and make design decisions.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Education Method</b>                 | The course Geometric Design of Roads and Railways is taught in lectures and practicum. The lectures explain the theory and principles of roads and railways design, and provide knowledge and insights on the fundamental issues in design. In the practicum (lab hours), students develop skills to apply this knowledge on real projects of designing roads and railways. The practicum is also meant to increase the understanding of the subject and to prepare the students for the assessment.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Assessment</b>                       | The assessment in this course is composed of the assessment of the two projects and the written examination (digital examination via ANS) as following:<br><br>o 35% First project (a minimum score of 5.0 is required)<br>o 25% Second project (a minimum score of 5.0 is required)<br>o 40% Written digital examination with registration on campus (a minimum score of 5.0 is required)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Expected prior Knowledge</b>         | CTB1420 - Transport & Planning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Academic Skills</b>                  | - Problem solving<br>- Critical thinking<br>- Design<br>- Reasoning<br>- Writing skills<br>- Presentation skills                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Literature &amp; Study Materials</b> | The following books and guidelines can be referred to:<br>Traffic & Highway Engineering, Fifth Edition, N.J. Garber, L.A. Hoel, 2009, CENGAGE Learning.<br>Designing Safe Road Systems: A Human Factors Perspective (Human Factors in Road and Rail Transport) by Jan Theeuwes, Richard Van Der Horst and Maria Kuiken (Aug 28, 2012).<br>NCHRP Report 600, Human Factors Guidelines for Road Systems, Second Edition, 2012.<br>Behavioural Adaptation and Road Safety: Theory, Evidence and Action by Christina Rudin-Brown and Samantha Jamson (May 24, 2013).<br>Modern Railway Track, Version 3.4, Coenraad Esveld, 2014.<br>ROA, Richtlijn Ontwerp Autosnelwegen 2019, Rijkswaterstaat, 2019 (The Dutch design guidelines for motorways).<br>CROW Handboek Wegontwerp (NL), (available from CROW kennisbank).<br>ASVV-Dutch guidelines in urban areas (NL), (available from CROW kennisbank).<br>CROW, Design manual for bicycle traffic (EN), June 2007.<br>CROW, Road safety manual (EN), May 2009.<br>Besides these references, specific material will also be provided during the lectures. |   |
| <b>Judgement</b>                        | For more information on grading, see article 14 in the Rules and Guidelines (RGE):<br><a href="https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations">https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Permitted Materials during Exam</b>  | No material is allowed during the exam.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Collegerama</b>                      | No                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |

| CTB3385                                 | Use of Underground Space                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 4 |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>           | Dr.ir. W. Broere                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Instructor</b>                       | Drs.ir. R.E.P. de Nijs                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Contact Hours / Week</b><br>x/x/x/x  | 0/0/4/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Education Period</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Start Education</b>                  | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Exam Period</b>                      | 3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Course Language</b>                  | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Expected prior knowledge</b>         | The course is open to students from other faculties and has no prerequisite courses. It is advisable to previously follow the courses Soil Mechanics and Design of Structures and Foundations.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Summary</b>                          | Students obtain basic knowledge of the multidisciplinary aspects of the use of underground space.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Course Contents</b>                  | Introduction and overview of the course<br>Geotechnical engineering (introduction)<br>Installation techniques and soil improvement methods<br>Legal aspects<br>Subsurface planning<br>Decision making process<br>Resilience & sustainability aspects<br>Basics of various kinds of tunnels<br>Small infrastructure<br>Basements<br>Tunnel cross sections & modalities<br>Tunnel safety and risk management<br>Social safety<br>Mixed use of land, layered use of land<br>New developments in construction techniques<br>Case 1 (actual project)<br>Case 2 (actual project)                                                                                                                                |   |
| <b>Study Goals</b>                      | explain characteristics of several construction technologies.<br>assess the construction technologies in different situations, with different geological or physical conditions.<br>analyse and structure the complex decision making process related to the use of underground space.<br>define an integral approach related to the use of underground space<br><br>Study load.<br>Lectures 14 x 2 hours: 28 hours<br>Hours related to lectures: 14 hours<br>Self-study (syllabus/sheets:) 25 hours<br>Paper: 35 hours<br>Exam preparation: 12 hours<br>Exam: 2 hour<br>Total 116 hours                                                                                                                  |   |
| <b>Education Method</b>                 | Lectures, written exam, writing a paper (group wise, minimum 2 maximum 4 students)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Literature and Study Materials</b>   | For the course, students can make use of the syllabus Use of Underground Space (190 pages) and the sheets of the lectures which are made available on BS. It is recommended but not compulsory that students also read the book Ondergronds bouwen (255 pages) COB (ISBN 90 807297 1 X).                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Assessment</b>                       | After the lectures students have to form groups consisting of two (minimum) to four (maximum) students and write a short paper (approx. 20 pages) about a topic than can either be chosen from a list of available topics or their own choice, and agreed upon by the lecturers. The topic has to be related to the use of underground space.<br><br>After the paper is finished a group wise mark is given for the paper, based on the coherence of the paper and the understanding and ability to apply the relevant information from the lectures, literature and syllabi.<br><br>A (computerized) written exam is taken individually.<br><br>The final mark is based on: 50% paper, 50% written exam. |   |
| <b>Permitted Materials during Tests</b> | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Contact</b>                          | The student-assistants of Underground Space Technology can be contacted at<br>E: StudassOGB-CiTG@tudelft.nl<br>T: 85256                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Expected prior Knowledge</b>         | The course is open to students from other faculties and has no prerequisite courses. It is advisable to previously follow the courses Soil Mechanics and Design of Structures and Foundations.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Academic Skills</b>                  | Knowledge on use of underground space, problem analysis, evaluation, literature review, scientific writing and reporting                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Literature &amp; Study Materials</b> | For the course, students can make use of the syllabus Use of Underground Space (190 pages) and the sheets of the lectures which are made available on BS. It is recommended but not compulsory that students also read the book Ondergronds bouwen (255 pages) COB (ISBN 90 807297 1 X).                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Judgement</b>                        | For more information on grading, see article 14 in the Rules and Guidelines (RGE):<br><a href="https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations">https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations</a>                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Permitted Materials during Exam</b>  | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Collegerama</b>                      | No                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |

| CTB3390                                 | Mechanics and Transport by Flow through Poreus Media                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 4 |
|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>           | S. Muraro                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Course Coordinator</b>               | S. Muraro                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Contact Hours / Week</b><br>x/x/x/x  | 0/0/4/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Education Period</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Start Education</b>                  | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Exam Period</b>                      | 3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Course Language</b>                  | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Course Contents</b>                  | <p>This course is jointly given with AESB3340.</p> <p>In this course, students acquire a fundamental basis for one- and two-phase flow in porous media. This course is an essential basis for all tracks within the MSc Applied Earth Sciences as well as the MSc track in Geotechnical Engineering in the MSc Civil Engineering. This course is highly recommended for students interested in Hydraulic Engineering as well as Hydro-geology and Water Management.</p> <p>The course will address the following topics:</p> <ul style="list-style-type: none"> <li>- basic principles for groundwater flow under saturated conditions (1D, 2D and 3D, both stationary as well as non-stationary flow);</li> <li>- (potential) energy and its relationship with pressure and head;</li> <li>- capillary pressure and unsaturated flow (2-phase flow in a porous medium filled with water and air);</li> <li>- concepts of water and air content, saturation, hydraulic conductivity, relative permeability, and residual saturation;</li> <li>- water retention curves (van Genuchten, Brooks-Corey, and Gardner equations);</li> <li>- the impact of flow on soil mechanics, highlighting the connection with the parallel course of soil mechanics.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Study Goals</b>                      | <p>At the end of this course you will be able to:</p> <ul style="list-style-type: none"> <li>- explain the origins of groundwater flow;</li> <li>- derive the governing equations of the groundwater flow with appropriate initial and boundary conditions for saturated soils and unsaturated soils;</li> <li>- solve laboratory and field steady-state and transient problems for saturated (1D, 2D) and unsaturated (1D) flow by programming a Python/Matlab script;</li> <li>- discuss the quality and the limitations of the model results.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Education Method</b>                 | <p>The course is structured along with weekly lectures and online assignments. The aim of this structure is to provide students with feedback on their progress. The content is structured with a study guide which is based on topics discussed in a number of standard handbooks.</p> <p>- Self-study<br/>The key to passing this course is self-study. The heart of the course is a study guide. In this guide, all topics are specified based on assignments that the students are required to make. The study guide points to the relevant chapters and paragraphs in a number of handbooks that the students need to study in order to be able to solve the assignments. There are several very good reference books available that cover the content of this course.</p> <p>- Lectures<br/>The students are expected to prepare for each lecture by making assignments and reading the required background material. The course is organised along with online assignments, activities integrated into Brightspace. These activities are a series of test assignments with which the students can evaluate their progress. Students are expected to upload questions to Brightspace about the assigned work or any subjects they have difficulties with. During the lectures, these questions will be addressed.</p> <p>- Asynchronous feedback and dialog<br/>Using the Discussion Board in the Brightspace environment, students are expected to collaborate in order to identify questions and to help each other complete the assignments. These tools are also used to give and get feedback on intermediate results (i.e. the assignments) but also on lectures etc.</p> |   |
| <b>Course Relations</b>                 | <p>This course is closely related with Soil Mechanics (CTB2310) and Physical Transport Phenomena (AESB2320). The course is fundamental to the Geo-engineering field and integrates different aspects with respect to the flow of water through porous media.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Literature and Study Materials</b>   | <p>Study guide, lecture notes, material on Brightspace, links to several good handbooks.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Assessment</b>                       | <p>Final exam in week 10. The exam consists of a series of questions that are similar to the assignments prepared for and discussed during the lectures.</p> <p>The online assignments given during the course are compulsory activities and enter in the final summative assessment (after the final exam is passed).</p> <p>The final grade is calculated as:<br/>70% exam + 30% Online Activities</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Permitted Materials during Tests</b> | <p>No intermediate tests will be given.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Contact</b>                          | <p>Dr. Stefano Muraro<br/>Email S.Muraro@tudelft.nl<br/>Phone +31 15 27 83327</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Expected prior Knowledge</b>         | <p>All Math, Mechanics 1, Soil Mechanics</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Academic Skills</b>                  | <p>Knowledge, students:</p> <ul style="list-style-type: none"> <li>- know the concepts and assumptions which form the basis of Darcy's Law;</li> <li>- are able to apply Darcy's law in a range of different applications;</li> <li>- understand how to simplify the 3D flow equation to 2D or 1D;</li> <li>- are able to explain when the assumptions used in the reduction of dimensionality are applicable;</li> <li>- understand the concepts of capillary pressure and unsaturated flow;</li> <li>- understand and be able to use the van Genuchten, Brooks-Corey, and Gardner equations;</li> </ul> <p>Skills, students:</p> <ul style="list-style-type: none"> <li>- are able to derive quantitative solutions to a range of engineering problems using analytical solutions;</li> <li>- have developed skills to share information and knowledge in an online environment;</li> <li>- have developed skills to help others in an online environment;</li> <li>- have developed the skill to use Python/Matlab to assess simple numerical and analytical solutions to flow and transport problems;</li> <li>- will have the skill to integrate transport of solutes with the groundwater flow problems in order to analyse groundwater contamination problems;</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                      |   |

|                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Literature &amp; Study Materials</b> | <p>Attitude, students:</p> <ul style="list-style-type: none"> <li>- will have developed an appreciation of the intrinsic value of the sub-surface for society, ecosystem etc.</li> <li>- will have developed an open attitude for collaboration in an online environment.</li> </ul> <p>All course materials are given on the Brightspace site.<br/>The course is based on the following books:<br/>Groundwater Science, 2013, Charles R. Fitts, second edition, Academic Press, Elsevier.<br/>Unsaturated Soil Mechanics, 2004, Lu, Ning; Likos, William J., John Wiley &amp; Sons.</p> <p>You are strongly advised to buy the book by Fitts. For Lu and Likos, we only use a small number of chapters. The relevant material will be provided during the class.</p> |
| <b>Judgement</b>                        | For more information on grading, see article 14 in the Rules and Guidelines (RGBE):<br><a href="https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations">https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations</a>                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Permitted Materials during Exam</b>  | The exam is an open-book exam, the student is allowed to bring all books, lecture notes, and worked examples. The student is allowed to use an electronic calculator. The exam is an individual assessment and no collaboration is allowed.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Collegerama</b>                      | No                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

| CTB3420                                 | Integral Design of Infrastructure                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 4 |
|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>           | Dr.ir. G.A. van Nederveen                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Instructor</b>                       | Dr. E.J. Houwing                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Contact Hours / Week x/x/x/x</b>     | 0/0/0/6<br>Hoorcollege 2 uur per week<br>Project: groepswork, 4 u/w begeleid                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Education Period</b>                 | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Start Education</b>                  | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Exam Period</b>                      | none                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Course Language</b>                  | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Parts</b>                            | <ul style="list-style-type: none"> <li>- Develop and execute a project plan;</li> <li>- Make a project analysis, including problem analysis, site analysis and stakeholder analysis and definition of the project scope;</li> <li>- Develop a design proposal; go through the full design cycle for a given case;</li> <li>- Realize the integration of the different engineering disciplines;</li> <li>- Elaborate specific parts or aspects;</li> <li>- Oral and written presentation of the design proposal.</li> </ul>                                                                                                                                                                                                                                  |   |
| <b>Summary</b>                          | Integration and application of different civil engineering disciplines in a design project setting.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Course Contents</b>                  | <p>Infrastructure projects are characterized by the complex setting in which they take place and the broad spectrum of involved technical disciplines: road, rail, structures, water, ground, etc. In other words: this kind of projects are multi-disciplinary and integral by definition. The planning and design process are executed in groups using a case assignment from practice. Systems engineering is an important method in this course.</p> <p>Subjects:</p> <ul style="list-style-type: none"> <li>- Project management,</li> <li>- Design approaches,</li> <li>- Structural engineering,</li> <li>- Systems engineering,</li> <li>- Economic and social aspects, sustainability, use and environment.</li> <li>- Risk management.</li> </ul> |   |
| <b>Study Goals</b>                      | <ul style="list-style-type: none"> <li>- Develop and execute a project plan;</li> <li>- Develop a design proposal; go through the full design cycle for a given case;</li> <li>- Realize the integration of the different engineering disciplines;</li> <li>- Elaborate specific parts or aspects;</li> <li>- Oral and written presentation of the design proposal.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Education Method</b>                 | Project course with supporting lectures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Course Relations</b>                 | <p>This course builds upon first year course CTB 1220 "Integraal Ontwerp en Beheer" and more generally upon knowledge gained in the first and second year of the Bachelor CT.</p> <p>This course uses and is relevant for all civil engineering disciplines.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Literature and Study Materials</b>   | Will be made available through BrightSpace                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Assessment</b>                       | Group reports (project plan, project analysis, design documents, final report), group presentations, and individual reports are assessed.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Expected prior Knowledge</b>         | Third year BSc Civil Engineering                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Academic Skills</b>                  | Collaboration in teams, presenting, reporting, information skills, awareness of societal impact of infrastructure projects                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Literature &amp; Study Materials</b> | Will be made available through Brightspace                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Judgement</b>                        | For more information on grading, see article 14 in the Rules and Guidelines (RGBE):<br><a href="https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations">https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations</a>                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Permitted Materials during Exam</b>  | n/a                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Collegerama</b>                      | No                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |

| CTB3425-17                              | Monitoring and Stability of Dikes and Embankments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 4 |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>           | Dr.ir. C. Zwanenburg                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Instructor</b>                       | Drs.ir. R.E.P. de Nijs                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Instructor</b>                       | Dr.ir. F.J. van Leijen                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Contact Hours / Week</b><br>x/x/x/x  | 0/0/0/4<br>20 lecture hours<br>12 tutorial hours<br>and 1 day excursion                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Education Period</b>                 | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Start Education</b>                  | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Exam Period</b>                      | 4<br>5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Course Language</b>                  | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Parts</b>                            | Lectures for principles and methodology<br>Tutorials for data elaboration and assessment of dikes and embankments<br>Analysis of a case history                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Summary</b>                          | The general learning goal of this course is to study basic geotechnical and geodetic aspects related to dikes and embankments, with special emphasis on monitoring and assessment of the stability and settlements. These skills should be applicable in a more general sense to the design, analysis, model, interpretation and visualisation in relation to geotechnical and geodetic problems.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Course Contents</b>                  | <p>Soil mechanics topics for dikes and embankments: Soils as construction materials. Volumetric behaviour of the soils; i.e. consolidation and creep, Soil shear strength, Seepage and free surface flow.</p> <p>Construction and lifetime behaviour: Generation and dissipation of pore pressures. Settlements and deformation. Geomechanical analysis of deformation. Influence of pore water pressure.</p> <p>Geotechnical monitoring: Measurement of displacements, pore pressure, water content. Design of a monitoring system for dikes, embankments and subsoil. Data elaboration and interpretation.</p> <p>Static 3D object reconstruction: Observation technologies for geometry Relevant estimation theory. Modelling 3D objects. Working with large data volumes. Visualisation in 3D.</p> <p>Dynamic 3D: Observation of deformation. Remote Sensing techniques, time series analysis, monitoring in (near) real-time.</p> <p>Observation and estimation theory: Aspects related to 3D dikes and embankments. Thresholds and approaches for critical situations.</p>                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Study Goals</b>                      | <p>At the end of this course you will be able to:</p> <ol style="list-style-type: none"> <li>1. list possible failure mechanisms of dikes and embankments and list their early indicators of possible failure.</li> <li>2. list the most important parameters to assess the stability of dikes and embankments.</li> <li>3. list and describe the element classes of visual inspection and provide a list of examples.</li> <li>4. describe the measurement principle, the error sources and the precision of different monitoring techniques.</li> <li>5. explain the geotechnical processes (consolidation, groundwater flow and instability) occurring in/under the dikes and embankments and apply corresponding models for assessment and predictions.</li> <li>6. carry out and assess a visual inspection of a dike.</li> <li>7. process various remote sensing and in-situ monitoring measurements, estimate optimal models, and interpret the results.</li> <li>8. process and analyze geotechnical and geodetic data in relation to deformation and stability, including quality assessment and control.</li> <li>9. assess displacements and stability of dikes and embankments subjected to hydro-mechanical loads.</li> <li>10. evaluate and summarize the results of observational data.</li> </ol> <p>design an optimal monitoring strategy for a certain situation, for instance the long-term monitoring of an embankment or a status check of a dike.</p> |   |
| <b>Education Method</b>                 | Lectures, tutorials, self-study, written reporting                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Course Relations</b>                 | Soil Mechanics<br>Survey and Mapping                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Assessment</b>                       | Final mark = final written online examination grade (60%) + evaluation of final report (40%)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Expected prior Knowledge</b>         | Soil Mechanics, Mathematics, knowledge of the concept of stress and strain as well as elasticity.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Academic Skills</b>                  | Critical thinking, Online learning, Note making, Time management, application of theory for problems, Basic mathematical skills                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Literature &amp; Study Materials</b> | <p>Course book: Soil Mechanics, A. Verruijt, Delft University of Technology, 2012. Go to: <a href="http://geo.verruijt.net/">http://geo.verruijt.net/</a> and download the book (in English) stored as SoilMechBook2012.pdf</p> <p>Accompanying lectures notes (in the form of pdf).</p> <p>Worked examples</p> <p>Lecture slides</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Judgement</b>                        | <p>For more information on grading, see article 14 in the Rules and Guidelines (RGBE): <a href="https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations">https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Permitted Materials during Exam</b>  | Calculator (see exam regulations for the allowed models)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Collegerama</b>                      | No                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |

H. Alkisaai

|                   |                   |
|-------------------|-------------------|
| <b>Department</b> | Applied Mechanics |
| <b>Department</b> | Applied Mechanics |
| <b>Room</b>       | 23.HG 6.47        |

J.A. Arriaga Garcia

|                   |                               |
|-------------------|-------------------------------|
| <b>Department</b> | Environmental Fluid Mechanics |
|-------------------|-------------------------------|

Dr.ir. J.M. Bloemendal

|                   |                           |
|-------------------|---------------------------|
| <b>Department</b> | Water Systems Engineering |
| <b>Telephone</b>  | +31 15 27 89916           |
| <b>Room</b>       | 23.HG 4.82                |

Dr.ir. T.S. van den Bremer

|                   |                               |
|-------------------|-------------------------------|
| <b>Department</b> | Environmental Fluid Mechanics |
| <b>Telephone</b>  | +31 15 27 83306               |
| <b>Room</b>       | 23.HG 2.87                    |

Dr.ir. W. Broere

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Civiele Techniek & Geowetensch |
| <b>Department</b> | Geo-engineering                |
| <b>Telephone</b>  | +31 15 27 81545                |
| <b>Room</b>       | 23.KG 00.510                   |

Dr.ir. N.P.K. Chan

|                   |              |
|-------------------|--------------|
| <b>Department</b> | Statistics   |
| <b>Room</b>       | 36.HB 06.130 |

Dr. C. Chassagne

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Civiele Techniek & Geowetensch |
| <b>Department</b> | Environmental Fluid Mechanics  |
| <b>Telephone</b>  | +31 15 27 85970                |
| <b>Room</b>       | 23.KG 03.120                   |

Dr.ir. A.M.J. Coenders

|                   |                                      |
|-------------------|--------------------------------------|
| <b>Unit</b>       | Civiele Techniek & Geowetensch       |
| <b>Department</b> | Water Systems Monitoring & Modelling |
| <b>Telephone</b>  | +31 15 27 82110                      |
| <b>Room</b>       | 23.HG 4.95                           |

Dr. O. Copuroglu

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Civiele Techniek & Geowetensch |
| <b>Department</b> | Materials and Environment      |
| <b>Telephone</b>  | +31 15 27 81827                |
| <b>Room</b>       | 23.HG 6.03                     |

J. Cupa

|                   |                   |
|-------------------|-------------------|
| <b>Department</b> | Applied Mechanics |
|-------------------|-------------------|

Dr.ir. K.N. van Dalen

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Civiele Techniek & Geowetensch |
| <b>Department</b> | Dynamics of Structures         |
| <b>Telephone</b>  | +31 15 27 88388                |
| <b>Room</b>       | 23.HG 3.61                     |

Prof.dr.ir. R.P.B.J. Dollevoet

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Civiele Techniek & Geowetensch |
| <b>Department</b> | Railway Engineering            |
| <b>Telephone</b>  | +31 15 27 82365                |
| <b>Room</b>       | 23.S2 2.28                     |



Dr.ir. R.J. van der Ent

|                   |                                      |
|-------------------|--------------------------------------|
| <b>Unit</b>       | Civiele Techniek & Geowetensch       |
| <b>Department</b> | Water Systems Monitoring & Modelling |
| <b>Telephone</b>  | +31 15 27 88327                      |
| <b>Room</b>       | 23.HG 4.92                           |

Prof.dr.ir. S.M.J.G. Erkens

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Civiele Techniek & Geowetensch |
| <b>Department</b> | Pavement Engineering           |
| <b>Telephone</b>  | +31 15 27 85949                |
| <b>Room</b>       | 23.S2 2.25                     |

Dr.ir. M.W. Ertsen

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Civiele Techniek & Geowetensch |
| <b>Department</b> | Water Systems Engineering      |
| <b>Telephone</b>  | +31 15 27 87423                |
| <b>Room</b>       | 23.HG 4.81                     |

Dr.ir. H. Farah

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Civiele Techniek & Geowetensch |
| <b>Department</b> | Traffic Systems Engineering    |
| <b>Telephone</b>  | +31 15 27 83401                |
| <b>Room</b>       | 23.HG 4.20                     |

L. Flessati

|                   |                 |
|-------------------|-----------------|
| <b>Department</b> | Geo-engineering |
| <b>Room</b>       | 23.KG 00.520    |

Prof.dr. K.G. Gavin

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Civiele Techniek & Geowetensch |
| <b>Department</b> | Geo-engineering                |
| <b>Telephone</b>  | +31 15 27 87128                |
| <b>Room</b>       | 23.KG 00.540                   |

Dr.ir. A. Gavriilidou

|                   |                             |
|-------------------|-----------------------------|
| <b>Department</b> | Traffic Systems Engineering |
| <b>Room</b>       | 23.HG 4.40                  |

Dr. rer. nat. G. George

|                   |                            |
|-------------------|----------------------------|
| <b>Department</b> | Atmospheric Remote Sensing |
|-------------------|----------------------------|

Prof.dr.ir. D. van Halem

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Civiele Techniek & Geowetensch |
| <b>Department</b> | Sanitary Engineering           |
| <b>Telephone</b>  | +31 15 27 85838                |
| <b>Room</b>       | 23.HG 4.48                     |

Prof.dr.ir. M.A.N. Hendriks

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Civiele Techniek & Geowetensch |
| <b>Department</b> | Engineering Structures         |
| <b>Telephone</b>  | +31 15 27 86988                |
| <b>Room</b>       | 23.S2 2.03                     |

Dr.ir. H. Hendrikse

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Civiele Techniek & Geowetensch |
| <b>Department</b> | Offshore Engineering           |
| <b>Telephone</b>  | +31 15 27 88223                |
| <b>Room</b>       | 23.HG 3.71                     |

Y. de las Heras

|                   |                                   |
|-------------------|-----------------------------------|
| <b>Unit</b>       | Civiele Techniek & Geowetensch    |
| <b>Department</b> | Education and Student Affairs CEG |
| <b>Telephone</b>  | +31 15 27 84800                   |

|             |            |
|-------------|------------|
| <b>Room</b> | 23.HG 2.73 |
|-------------|------------|

## Prof.dr.ir. M.J.C.M. Hertogh

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Civiele Techniek & Geowetensch |
| <b>Department</b> | Integral Design & Management   |
| <b>Telephone</b>  | +31 15 27 84921                |
| <b>Room</b>       | 23.HG 6.63                     |

## Prof.dr. M.A. Hicks

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Civiele Techniek & Geowetensch |
| <b>Department</b> | Geo-engineering                |
| <b>Telephone</b>  | +31 15 27 87433                |
| <b>Room</b>       | 23.KG 00.060                   |

## Dr.ir. O.A.C. Hoes

|                   |                                      |
|-------------------|--------------------------------------|
| <b>Unit</b>       | Civiele Techniek & Geowetensch       |
| <b>Department</b> | Water Systems Monitoring & Modelling |
| <b>Telephone</b>  | +31 15 27 84772                      |
| <b>Room</b>       | 23.HG 4.90                           |

## Dr. E.J. Houwing

|                   |                              |
|-------------------|------------------------------|
| <b>Department</b> | Integral Design & Management |
| <b>Room</b>       | 23.HG 6.71                   |

## Dr.ir. R.W. Hut

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Civiele Techniek & Geowetensch |
| <b>Department</b> | Water Resources                |
| <b>Telephone</b>  | +31 15 27 84909                |
| <b>Room</b>       | 23.HG 4.76                     |

## A. Jagadeesh

|                   |                      |
|-------------------|----------------------|
| <b>Department</b> | Pavement Engineering |
|-------------------|----------------------|

## Dr. O. Jaibi

|                   |                      |
|-------------------|----------------------|
| <b>Department</b> | Mathematical Physics |
| <b>Telephone</b>  | +31 15 27 88548      |

## Prof.dr. H.M. Jonkers

|                   |                                           |
|-------------------|-------------------------------------------|
| <b>Unit</b>       | Civiele Techniek & Geowetensch            |
| <b>Department</b> | Materials- Mechanics- Management & Design |
| <b>Telephone</b>  | +31 15 27 82313                           |
| <b>Room</b>       | 23.HG 6.07                                |

## Prof.dr. C.A. Katsman

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Civiele Techniek & Geowetensch |
| <b>Department</b> | Environmental Fluid Mechanics  |
| <b>Telephone</b>  | +31 15 27 83380                |
| <b>Room</b>       | 23.HG 2.89                     |

## Dr. F. Kavoura

|                   |                              |
|-------------------|------------------------------|
| <b>Department</b> | Steel & Composite Structures |
| <b>Room</b>       | 23.S2 2.54                   |

## Dr.ir. M. Keijzer

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Elektrotechn., Wisk. & Inform. |
| <b>Department</b> | Mathematical Physics           |
| <b>Telephone</b>  | +31 15 27 85803                |
| <b>Room</b>       | 36.HB 05.240                   |

## S.J. Kestens

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Civiele Techniek & Geowetensch |
| <b>Department</b> | Management Support             |

|                           |                               |
|---------------------------|-------------------------------|
| <b>Telephone<br/>Room</b> | +31 15 27 87515<br>23.HG 4.52 |
|---------------------------|-------------------------------|

---

Dr. V.L. Knoop

|                            |                                                               |
|----------------------------|---------------------------------------------------------------|
| <b>Unit<br/>Department</b> | Civiele Techniek & Geowetensch<br>Traffic Systems Engineering |
| <b>Telephone<br/>Room</b>  | +31 15 27 88413<br>23.HG 4.37                                 |

---

Dr.ir. M. Korff

|                            |                                                   |
|----------------------------|---------------------------------------------------|
| <b>Unit<br/>Department</b> | Civiele Techniek & Geowetensch<br>Geo-engineering |
| <b>Room</b>                | 23.KG 00.140                                      |

---

Prof.dr.ir. M.K. de Kreuk

|                            |                                                    |
|----------------------------|----------------------------------------------------|
| <b>Unit<br/>Department</b> | Civiele Techniek & Geowetensch<br>Water Management |
| <b>Telephone</b>           | +31 15 27 85274                                    |

---

Dr.ir. R.J. Labeur

|                            |                                                                 |
|----------------------------|-----------------------------------------------------------------|
| <b>Unit<br/>Department</b> | Civiele Techniek & Geowetensch<br>Environmental Fluid Mechanics |
| <b>Telephone<br/>Room</b>  | +31 15 27 85069<br>23.HG 2.94                                   |

---

Dr.ir. F.J. van Leijen

|                            |                                                                        |
|----------------------------|------------------------------------------------------------------------|
| <b>Unit<br/>Department</b> | Civiele Techniek & Geowetensch<br>Mathematical Geodesy and Positioning |
| <b>Telephone<br/>Room</b>  | +31 15 27 85719<br>23.HG 2.29                                          |

---

Dr. S.L.M. Lhermitte

|                            |                                                                        |
|----------------------------|------------------------------------------------------------------------|
| <b>Unit<br/>Department</b> | Civiele Techniek & Geowetensch<br>Mathematical Geodesy and Positioning |
| <b>Telephone<br/>Room</b>  | +31 15 27 85955<br>23.HG 2.22                                          |

---

Prof.dr. Z. Li

|                            |                                                       |
|----------------------------|-------------------------------------------------------|
| <b>Unit<br/>Department</b> | Civiele Techniek & Geowetensch<br>Railway Engineering |
| <b>Telephone<br/>Room</b>  | +31 15 27 82325<br>23.S2 2.29                         |

---

Prof.dr.ir. J.B. van Lier

|                            |                                                        |
|----------------------------|--------------------------------------------------------|
| <b>Unit<br/>Department</b> | Civiele Techniek & Geowetensch<br>Sanitary Engineering |
| <b>Telephone<br/>Room</b>  | +31 15 27 81615<br>23.HG 4.57                          |

---

Dr. M. Lukovi

|                            |                                                       |
|----------------------------|-------------------------------------------------------|
| <b>Unit<br/>Department</b> | Civiele Techniek & Geowetensch<br>Concrete Structures |
| <b>Telephone<br/>Room</b>  | +31 15 27 82320<br>23.S2 1.04                         |

---

Dr.ir. H. van der Marel

|                            |                                                                        |
|----------------------------|------------------------------------------------------------------------|
| <b>Unit<br/>Department</b> | Civiele Techniek & Geowetensch<br>Mathematical Geodesy and Positioning |
| <b>Telephone<br/>Room</b>  | +31 15 27 84907<br>23.HG 2.20                                          |

---

Dr. V.L. Markine

|                            |                                                       |
|----------------------------|-------------------------------------------------------|
| <b>Unit<br/>Department</b> | Civiele Techniek & Geowetensch<br>Railway Engineering |
|----------------------------|-------------------------------------------------------|

|                  |                 |
|------------------|-----------------|
| <b>Telephone</b> | +31 15 27 83206 |
| <b>Room</b>      | 23.S2 2.33      |

---

## Dr. E. Mostert

|                   |                                   |
|-------------------|-----------------------------------|
| <b>Unit</b>       | Civiele Techniek & Geowetensch    |
| <b>Department</b> | Surface and Groundwater Hydrology |
| <b>Telephone</b>  | +31 15 27 87800                   |
| <b>Room</b>       | 23.HG 4.82                        |

---

## S. Muraro

|                   |                 |
|-------------------|-----------------|
| <b>Department</b> | Geo-engineering |
| <b>Room</b>       | 23.KG 00.510    |

---

## Dr.ir. G.A. van Nederveen

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Civiele Techniek & Geowetensch |
| <b>Department</b> | Integral Design & Management   |
| <b>Telephone</b>  | +31 15 27 81854                |
| <b>Room</b>       | 23.HG 6.64                     |

---

## Drs.ir. R.E.P. de Nijs

|                   |                 |
|-------------------|-----------------|
| <b>Department</b> | Geo-engineering |
| <b>Room</b>       | 23.KG 00.510    |

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Civiele Techniek & Geowetensch |
| <b>Department</b> | Geo-engineering                |
| <b>Room</b>       | 23.KG 00.510                   |

---

## Dr. A.A. Nuijens

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Civiele Techniek & Geowetensch |
| <b>Department</b> | Atmospheric Remote Sensing     |
| <b>Telephone</b>  | +31 15 27 85956                |
| <b>Room</b>       | 23.HG 2.22                     |

---

## Dr.ir. N. van Oort

|                   |                                   |
|-------------------|-----------------------------------|
| <b>Unit</b>       | Civiele Techniek & Geowetensch    |
| <b>Department</b> | Transport, Mobility and Logistics |
| <b>Room</b>       | 23.HG 4.10.2                      |

---

## Dr.ir. D. den Ouden-van der Horst

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Elektrotechn., Wisk. & Inform. |
| <b>Department</b> | Numerical Analysis             |
| <b>Room</b>       | 36.HB 03.250                   |

---

## R. Oval

|                   |                   |
|-------------------|-------------------|
| <b>Department</b> | Applied Mechanics |
|-------------------|-------------------|

|                   |                   |
|-------------------|-------------------|
| <b>Department</b> | Applied Mechanics |
|-------------------|-------------------|

|                   |                   |
|-------------------|-------------------|
| <b>Department</b> | Applied Mechanics |
|-------------------|-------------------|

---

## Ir. S. Pasterkamp

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Civiele Techniek & Geowetensch |
| <b>Department</b> | Applied Mechanics              |
| <b>Telephone</b>  | +31 15 27 84982                |
| <b>Room</b>       | 23.HG 6.46                     |

---

## Dr. M. Pavlovic

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Civiele Techniek & Geowetensch |
| <b>Department</b> | Steel & Composite Structures   |
| <b>Telephone</b>  | +31 15 27 83382                |
| <b>Room</b>       | 23.S2 2.58                     |

Dr.ir. G.J.P. Ravenshorst

|                   |                                  |
|-------------------|----------------------------------|
| <b>Unit</b>       | Civiele Techniek & Geowetensch   |
| <b>Department</b> | Bio-based Structures & Materials |
| <b>Telephone</b>  | +31 15 27 85721                  |
| <b>Room</b>       | 23.S2 2.57                       |

R.H.C. Reudink

|                   |                                          |
|-------------------|------------------------------------------|
| <b>Unit</b>       | Civiele Techniek & Geowetensch           |
| <b>Department</b> | Laboratory Geoscience and Remote Sensing |
| <b>Telephone</b>  | +31 15 27 85728                          |
| <b>Room</b>       | 23.HG 2.01                               |

Prof.dr.ir. L.C. Rietveld

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Civiele Techniek & Geowetensch |
| <b>Department</b> | Sanitary Engineering           |
| <b>Telephone</b>  | +31 15 27 84732                |
| <b>Room</b>       | 23.HG 4.53                     |

Prof.dr.ir. J.G. Rots

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Civiele Techniek & Geowetensch |
| <b>Department</b> | Applied Mechanics              |
| <b>Telephone</b>  | +31 15 27 83799                |
| <b>Room</b>       | 23.HG 6.71                     |

Dr.ir. A.M. Salomons

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Civiele Techniek & Geowetensch |
| <b>Department</b> | Traffic Systems Engineering    |
| <b>Telephone</b>  | +31 15 27 88556                |
| <b>Room</b>       | 23.HG 4.45                     |

Dr.ir. H.R. Schipper

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Civiele Techniek & Geowetensch |
| <b>Department</b> | Applied Mechanics              |
| <b>Telephone</b>  | +31 15 27 89933                |

Dr.ir. G.H.W. Schoups

|                   |                                   |
|-------------------|-----------------------------------|
| <b>Unit</b>       | Civiele Techniek & Geowetensch    |
| <b>Department</b> | Surface and Groundwater Hydrology |
| <b>Telephone</b>  | +31 15 27 84508                   |
| <b>Room</b>       | 23.HG 4.94                        |

A.C.B. Schuurman

|                   |                   |
|-------------------|-------------------|
| <b>Department</b> | Applied Mechanics |
| <b>Department</b> | Applied Mechanics |
| <b>Room</b>       | 23.HG 6.48        |

Dr. C.O. Smet

|                   |                 |
|-------------------|-----------------|
| <b>Department</b> | Statistics      |
| <b>Telephone</b>  | +31 15 27 88666 |
| <b>Room</b>       | 36.HB 06.260    |

Dr.ir. M.J.M.M. Steenbergen

|                   |                                     |
|-------------------|-------------------------------------|
| <b>Unit</b>       | Civiele Techniek & Geowetensch      |
| <b>Department</b> | Mechanics and Physics of Structures |
| <b>Telephone</b>  | +31 15 27 83385                     |
| <b>Room</b>       | 23.S2 2.32                          |

Dr.ing. K.C. Terwel

|                   |                                |
|-------------------|--------------------------------|
| <b>Department</b> | Applied Mechanics              |
| <b>Telephone</b>  | +31 15 27 81512                |
| <b>Unit</b>       | Civiele Techniek & Geowetensch |

|                   |                   |
|-------------------|-------------------|
| <b>Department</b> | Applied Mechanics |
| <b>Telephone</b>  | +31 15 27 81512   |
| <b>Room</b>       | 23.HG 6.46        |

---

Dr.ir. C.C.J.M. Tiberius

|                   |                                      |
|-------------------|--------------------------------------|
| <b>Unit</b>       | Civiele Techniek & Geowetensch       |
| <b>Department</b> | Mathematical Geodesy and Positioning |
| <b>Telephone</b>  | +31 15 27 82573                      |
| <b>Room</b>       | 23.HG 2.23                           |

---

Prof.dr.ir. R. Uijlenhoet

|                   |                                      |
|-------------------|--------------------------------------|
| <b>Department</b> | Water Systems Monitoring & Modelling |
| <b>Room</b>       | 23.HG 4.87                           |

---

Prof.dr. P.J. Vardon

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Civiele Techniek & Geowetensch |
| <b>Department</b> | Geo-engineering                |
| <b>Telephone</b>  | +31 15 27 81456                |
| <b>Room</b>       | 23.KG 00.490                   |

---

Dr.ing. M.Z. Voorendt

|                   |                                     |
|-------------------|-------------------------------------|
| <b>Unit</b>       | Civiele Techniek & Geowetensch      |
| <b>Department</b> | Hydraulic Structures and Flood Risk |
| <b>Telephone</b>  | +31 15 27 83340                     |
| <b>Room</b>       | 23.HG 3.76                          |

---

Ir. P.A. de Vries

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Civiele Techniek & Geowetensch |
| <b>Department</b> | Macromechanics Laboratory      |
| <b>Telephone</b>  | +31 15 27 84034                |
| <b>Room</b>       | 23.S2 2.56                     |

---

Prof.dr.ir. C. Vuik

|                   |                                        |
|-------------------|----------------------------------------|
| <b>Unit</b>       | Elektrotechn., Wisk. & Inform.         |
| <b>Department</b> | Delft Institute of Applied Mathematics |
| <b>Telephone</b>  | +31 15 27 85530                        |
| <b>Room</b>       | 36.HB 03.320                           |

---

Ir. J.W. Welleman

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Civiele Techniek & Geowetensch |
| <b>Department</b> | Applied Mechanics              |
| <b>Telephone</b>  | +31 15 27 84856                |
| <b>Room</b>       | 23.HG 6.55                     |

---

Dr.ir. J.A. van der Werf

|                   |                           |
|-------------------|---------------------------|
| <b>Department</b> | Water Systems Engineering |
| <b>Telephone</b>  | +31 15 27 82258           |
| <b>Room</b>       | 23.HG 4.64                |

---

|                   |                      |
|-------------------|----------------------|
| <b>Department</b> | Sanitary Engineering |
| <b>Telephone</b>  | +31 15 27 82258      |
| <b>Room</b>       | 23.HG 4.64           |

---

|                   |                      |
|-------------------|----------------------|
| <b>Department</b> | Sanitary Engineering |
| <b>Telephone</b>  | +31 15 27 82258      |
| <b>Room</b>       | 23.HG 4.64           |

---

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Civiele Techniek & Geowetensch |
| <b>Department</b> | Sanitary Engineering           |
| <b>Telephone</b>  | +31 15 27 82258                |
| <b>Room</b>       | 23.HG 4.64                     |

---

T.R. van Woudenberg

|                   |                   |
|-------------------|-------------------|
| <b>Department</b> | Applied Mechanics |
| <b>Telephone</b>  | +31 15 27 89739   |

|             |            |
|-------------|------------|
| <b>Room</b> | 23.HG 6.45 |
|-------------|------------|

M. Zegel

|                   |                          |
|-------------------|--------------------------|
| <b>Department</b> | Support Water Management |
| <b>Room</b>       | 23.HG 4.52               |

Ir. T.J. Zitman

|                   |                                |
|-------------------|--------------------------------|
| <b>Department</b> | Coastal Engineering            |
| <b>Telephone</b>  | +31 15 27 84739                |
| <b>Department</b> | Coastal Engineering            |
| <b>Telephone</b>  | +31 15 27 84739                |
| <b>Unit</b>       | Civiele Techniek & Geowetensch |
| <b>Department</b> | Coastal Engineering            |
| <b>Telephone</b>  | +31 15 27 84739                |
| <b>Room</b>       | 23.HG 3.90                     |

Dr.ir. C. Zwanenburg

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Civiele Techniek & Geowetensch |
| <b>Department</b> | Geo-engineering                |
| <b>Room</b>       | 23.KG 00.500                   |